# Similarity and social discounting: Measuring preferences for social distance*

Geoffrey Castillo† Benjamin Beranek‡

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Altruism is typically assumed to decline with social distance. We test this assumption by estimating the shape of individual social discount functions. In two experiments, we manipulate social distance continuously by matching subjects with recipients who vary in how many demographic characteristics are shared between them. While many subjects are social discounters and favour closer recipients, we find that a quarter to a third of subjects are 'social magnifiers' and favour more distant ones. Considering choices between $1 for a distant recipient and money for themselves, our estimates imply that social discounters would accept a discount of $0.24 to have it redirected to themselves, while social magnifiers would require a premium of $0.11. Social discounting and magnification remain after taking self bias into account. We further find that many subjects have non-monotone preferences.

**Keywords:** similarity, social distance, social discounting, choice experiment, Inclusion of Other in the Self
**JEL Codes:** C91, D64, D91

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†Corresponding author. Department of Economics, Nottingham Business School, Nottingham Trent University. `geoffrey.castillo@ntu.ac.uk`

‡Department of Finance, Economics and Risk Management, Missouri State University. `benjaminberanek@missouristate.edu`

# 1 Introduction

People are commonly thought to be more altruistic towards those who are socially closer to them. We test this directly by estimating how altruism varies over a continuous measure of social distance. We find that this assumption does not always hold and that there is considerable heterogeneity in preferences for social distance.

To capture preferences for social distance, we start by setting up a structural model. The model draws from a long tradition in economics which views social distance as an important attribute and suggests that people discount payoffs as a function of social distance (Edgeworth, 1881; Marshall, 1975; Smith, 1759).[1] The model also draws from the literature in psychology which coined the term 'social discounting' (Jones and Rachlin, 2006).

In the model, a decision-maker faces options $(x, s)$. Attribute $x$ is an amount of money allocated to a recipient while attribute $s$ is the social distance between the decision-maker and the recipient of $x$. Preferences over these options are captured by the utility function $U(x, s) = D(s)\, u(x)$, where $D(s)$ is the social discount function and $u(x)$ is the utility of money.

The central objective of this paper is to determine the shape of the social discount function $D(s)$. A large literature, which we review in the next section, suggests that $D(s)$ is downward-sloping: payoffs are discounted as social distance increases. However, most studies focus on binary social distances, for example by comparing altruism towards identifiable and non-identifiable recipients, ingroup and outgroup members, or friends and strangers.[2] To precisely estimate the social discount function $D(s)$ over its whole support, we need to use a continuous measure of social distance $s$.

Our continuous measure of social distance is interpersonal similarity; that is, the number of demographic characteristics shared between people. Similarity is a major component of social distance (Karakayali, 2009; Liviatan et al., 2008; Rummel, 1976) and is by itself an important concept: it has been linked to positive outcomes in many domains, for instance in the labour market, health, trade, and education.[3] We devise an experimental method that allows us to manipulate similarity—and through it, social distance—by pairing subjects with multiple others who are more or less similar to them. Using similarity over other methods allows us to manipulate social distance without additional measurements

[1] As highlighted by Collard (1975) and Sally (2001).

[2] See for example, and respectively, Bohnet and Frey (1999a,b), Chen and Li (2009), and Leider et al. (2009). We review this literature in greater detail in the next section.

[3] See respectively Behncke et al. (2010), Alsan et al. (2019), Guiso et al. (2009), and Gershenson et al. (2022). We come back to this literature in the next section.

beyond demographic information. It also allows us to manipulate social distance without relying on existing relationships, which controls for reciprocity and reputation concerns.

In the experiment, we present subjects with a series of binary choices between options $(x, s)$. We systematically vary both the amount of money $x$ and similarity $s$ to generate many options and thus many choices. We then estimate each subject's social discount function $D(s)$ nonparametrically which reveals its shape with minimal assumptions. We also assume parametric forms for $D(s)$ and estimate its structural parameters. For example, we use the exponential social discount function $D(s) = \delta^s$ and estimate $\delta$, the social discount factor. We use a random coefficient framework to capture heterogeneity in preferences: we assume that parameters are drawn from a distribution and we estimate the parameters of this distribution (Conte et al., 2011; von Gaudecker et al., 2011).

In line with the literature, we find that most people are more altruistic towards those who are socially closer to them. These 'social discounters' have a downward-sloping social discount function $D(s)$, meaning that they discount payoffs given to others more as social distance increases. In terms of the structural model, this is reflected by a social discount factor $\delta$ between 0 and 1. Our estimates imply that social discounters have a dissimilarity premium of about \$0.24 to the dollar, meaning that, on average, they would be willing to forgo up to \$0.24 of a dollar to have it redirected to themselves—and thus would be willing to accept as little as \$0.76—rather than letting a socially distant other have it.

However, and in contrast to the literature, we find considerable heterogeneity in how social distance affects preferences. In particular, our analysis reveals that a quarter to a third of people are more altruistic towards those who are socially farther. These 'social magnifiers' are captured by an upward-sloping social discount function. Their dissimilarity premium is negative, at about $-$\$0.11, meaning that, on average, they would rather let a socially distant other have \$1 than receive up to \$1.11 themselves. We also find that social distance preferences are not necessarily monotone with some people preferring closer and more distant others to those in between or even the opposite.

We devise an extension of our experiment in which we include self as a potential recipient of the allocations. This modification allows us to test for hyperbolic discounting. We conduct this extension on a separate sample and estimate a quasi-hyperbolic social discount function $D(s) = \beta\delta^s$ for $s > 0$, in which $\beta$ captures 'self bias'. This function parallels the quasi-hyperbolic model in intertemporal choice (Laibson, 1997). In addition to replicating our original results, we find evidence for quasi-hyperbolic social discounting. The dissimilarity premium rises to \$0.62 for social discounters and to \$0.38 for social magnifiers, reflecting a substantial additional bias toward the self.

We then assess the robustness of our findings by considering alternative explanations.

We use a separate measure of social closeness—the Inclusion of Other in the Self (IOS, Aron et al., 1992) scale—to make sure that the results we find using interpersonal similarity hold when we instead order recipients according to their perceived closeness. The IOS scale is one of the most popular measures of social closeness in social psychology and is now routinely used in economics.[4] We test whether redistributive concerns explain away our results by controlling for the income of the recipient. We consider alternative functional forms and distributional assumptions. All of these additional analyses confirm our main findings.

Finally, we test whether our results extend beyond our setting by re-analysing data from a lab-in-the-field experiment in Uganda (Robson, 2021). It differs from ours in its subject population and its task. Subjects were also never shown the demographic characteristics of their recipients and could only infer them by seeing each other, which addresses the concern that our results are an artefact of providing this information. Social magnifiers and non-monotone preferences are even more common in these data than in ours.

Our findings matter for how one models, measures, and studies the effect of social distance on altruism. Models typically assume that altruism declines with social distance (e.g., Tabellini, 2008). Our results show that this assumption fails for a sizeable minority. As a result, the social discount function should not be assumed to be downward-sloping or monotone, at least not for everyone. Further, we provide a method to measure continuous social distance via similarity that can be easily integrated into other designs and other studies. This method already addresses an identified need in the literature. To echo Enke (2024): "a key unobservable is the notion of 'social distance' .... A potentially promising approach is that of similarity: social psychologists highlight that people generally feel closer to others that they perceive to be similar to them. This suggests conceptualizing social distance as perceived dissimilarity. Building on this idea, one could imagine devising experimental or survey techniques that measure people's perceived similarity to different groups, and to link behavior ... to these similarity judgments." This is exactly what this paper does. Finally, our results suggest that policies that impact others at different social distances may not be perceived in the same way by everyone. For example, some may not support a redistributive policy that benefits socially distant people, while others may be enthusiastic about it.

[4]See Aron et al. (2013), Branand et al. (2019) and Aron et al. (2022) for reviews, and Gächter et al. (2015) for a recent validation. On its usage in economics, see Castillo (2021), Dimant (2024), Goette and Tripodi (2021), Hofmann et al. (2021), Hofmann (2025), Robson (2021), Schütt (2023), and Gächter et al. (2025) for recent examples.

We conclude the paper with a discussion of our results. In particular, we highlight a potential connection between social distance preferences and time preferences.

## 2 Related literature

A large and diverse literature has studied the effect of social distance on altruism. We make three contributions to this literature: we rely on a continuous measure of similarity to manipulate social distance, we estimate a structural model based on social discounting, and we reconsider the finding of altruism towards socially closer others.

The methods used in this literature to manipulate social distance can be divided into three broad categories. A first category manipulates the feeling of closeness by increasing the identifiability of the recipient (Bohnet and Frey, 1999a; Buchan et al., 2006; Charness and Gneezy, 2008; Dufwenberg and Muren, 2006).[5] Charness and Gneezy (2008), for example, reveal the name of the recipient. A second category relies on the minimal group paradigm (Balliet et al., 2014; Charness and Chen, 2020; Chen and Li, 2009; Lane, 2016). For instance, Chen and Li (2009) divide subjects into two groups based on their preferences for paintings by Klee or Kandinsky. A third category brings existing relationships into the experiment (Gächter et al., 2022, 2025; Glaeser et al., 2000), a subset of which further elicits the full social network (Brañas-Garza et al., 2010; Goeree et al., 2010; Leider et al., 2009). Goeree et al. (2010), for example, elicit the network of friendships of 10- to 18-year-old students in a school.

Our method to manipulate social distance sits at the intersection of these methods. Instead of a binary social distance—identifiable vs non-identifiable, ingroup vs outgroup, known vs unknown others—we use similarity to generate continuous social distance.[6] Being continuous, it accounts for the fact that people belong to multiple social groups and that, when considering others, some group memberships overlap and others do not. Conveniently for an experimenter, similarity does not require an extra measurement beyond demographic information which is often available in administrative data or included in the standard questionnaires they use already. Finally, using similarity maintains full

[5] A related but distinct literature manipulates the social distance between subject and experimenter by varying the anonymity of the giver themselves, for example by comparing single-blind and double-blind procedures (Bohnet and Frey, 1999b; Hoffman et al., 1996, 1999).

[6] In studies that elicit a social network (Brañas-Garza et al., 2010; Goeree et al., 2010; Leider et al., 2009), social distance is defined as the shortest path on the network. For example, a direct friend receives a distance of one, a friend of a friend a distance of two, and so on. Instead of studying the compounded effect of links, we study the strength of a single link as captured by similarity. Our method could be combined with this approach to allow for weighting the links in the social network by similarity.

anonymity since it does not require existing relationships. It thus controls for reciprocity and reputation concerns, which have been shown to be important motives for sharing (Ligon and Schechter, 2012).

In terms of the analysis, most of the literature has used reduced-form models or models that capture the effect of social distance with a binary variable. We build on this literature by using a structural model explicitly based on social discounting. Many previous approaches can be viewed as special cases or reduced-form approximations of this model. The structural model allows us to estimate deep preference parameters each with a clear interpretation. With the model, we can test different functional forms for the social discount function such as the quasi-hyperbolic social discount function. We can bring in concepts such as similarity equivalent and dissimilarity premium that are parallel to the concepts of certainty equivalent and risk premium in the context of risk preferences. Finally, we can control for the curvature of the utility function of money which can impact the estimate of the social discount factor.

In terms of results, we find that a majority of our subjects are more altruistic towards socially closer others. This finding aligns with the literature. For example, Charness and Gneezy (2008) find that subjects allocate more when the name of the recipient is revealed; Chen and Li (2009), when the recipient is in the ingroup; and Goeree et al. (2010), when the recipient is closer on the social network. However, we also find that about a third of subjects are more altruistic towards socially farther others and that some subjects have non-monotone preferences for social distance. In addition, we demonstrate existence of a self bias when using a quasi-hyperbolic social discount function.

The two closest papers to ours are Schütt (2023) and Robson (2021), the former because of the method used to manipulate social distance and the latter because of the use of a structural model. Schütt (2023) also uses similarity to manipulate social distance: it administers a personality test and matches subjects based on the personality scores. The scores and matching procedure are revealed to subjects. In contrast, we use demographic characteristics to measure similarity, and we do not reveal to subjects how they have been matched. In addition, we use a structural model to analyse subjects' choices. Robson (2021) uses a CES utility function as a structural model to capture how social closeness (measured with the IOS scale) affects altruism. The use of social discounting allows us to study the social discount function separately and estimate its parameters. We give more details on the differences between our study and Robson (2021) when we re-analyse its data in Section 7.3.

The term 'social discounting' was coined in psychology by Jones and Rachlin (2006).[7] Compared to this literature, we make the same three contributions mentioned above. The vast majority of studies on social discounting in psychology use hypothetical social distances and thus hypothetical choices.[8] For example, subjects are asked to imagine a list of 100 people, ranging from the socially closest at position #1 to the most distant at #100. Then, they face a matching task: they consider options of the type 'A. \$155 for you alone. B. \$75 for you and 75 for the #[ ] person on the list' and they have to fill the blank to reach indifference between the two options. Data from these decisions is used to estimate social discount functions while assuming a linear utility function. Compared to this, we use real social distances and real choices while still keeping complete anonymity between subjects and recipients. We use a choice task instead of a matching task, and we also take into account the curvature of the utility function. Both of these elements have been shown to be important in the context of the elicitation of time preferences (Frederick et al., 2002). Finally, in terms of the results, the social discounting literature finds that all subjects are social discounters. As we have already mentioned, we do not find this to be the case.

Another closely related literature is that on moral universalism, which measures the extent to which people's altruism is invariant to the identity or group membership of the recipient (Cappelen et al., 2025; Enke, 2024; Enke et al., 2022, 2023). In this literature, subjects split a hypothetical amount of money between two recipients at different social distances: a randomly selected member of one of their social groups and a stranger from their country ('domestic' universalism), a compatriot and a world citizen ('foreign'), or two world citizens, one of whom shares a social group with them ('global'). As in our original design, subjects never allocate money to themselves.

Compared to our study, the same three key differences as above apply to the moral universalism literature. First, social distance is assumed rather than measured in this literature: each decision contrasts two recipients whose relative position on the social distance scale is taken as given rather than elicited. Enke (2024) identifies this as the literature's main open conceptual question and calls for techniques that measure perceived similarity to different groups and link it to behaviour in the allocation tasks. Our measure

[7] See also Rachlin and Jones (2008a,b) as well as Jones (2022) for a recent review. Social discounting has been applied to several topics, for example to explain contributions in the public goods game (Jones and Rachlin, 2009), cigarette smoking during pregnancy (Bradstreet et al., 2012), externalising behaviour problems (Sharp et al., 2012), and organ donation (Vekaria et al., 2017).

[8] An exception is Locey et al. (2011) who asked subjects to bring the names and address of people at positions #1, #2, #5, #10, and #20. Then, the questions continued to refer to these positions. Actual choices were implemented if people at these positions were selected for payment.

of similarity does this, and we show that it moves perceived closeness. Second, these tasks measure universalism as the share of the budget given to the more distant recipient averaged across decisions. Each subject is therefore summarised by a single number, so heterogeneity can only be in how universalist someone is and not in the shape of their social discount function. The structural model we use lets us estimate the shape of the social discount function $D(s)$ over its whole range. Third, subjects who favour the socially distant recipient are rare in these data.[9] In contrast, we find that about a third of our subjects are more altruistic towards socially distant others and that some have non-monotone preferences for social distance. These differences reflect different objectives: the moral universalism literature builds a measure that can be deployed at scale to explain outcomes such as donations, home bias, or political views, while we aim to structurally estimate the social discount function with a controlled experiment.

Finally, we relate to a large literature that studies the impact of similarity on economic outcomes. For example, Behncke et al. (2010) show that unemployed individuals are more likely to find and retain a job when they are matched with caseworkers who are of the same age, gender, education, and nationality. Alsan et al. (2019) show that Black patients are more likely to choose preventive health interventions when they are treated by Black doctors. Other examples include the perception of migrants, education, international trade, development, and sports.[10] Similarity can drive these results through a variety of channels. For example, Behncke et al. (2010) find that people trust each other more as their similarity increases, and Alsan et al. (2019) find evidence that similarity leads to better communication. We contribute to this literature by isolating the effect of similarity. We show that even after removing competing channels—trust, communication, more frequent interactions, and so on—similarity still has an effect on behaviour.

[9] Enke et al. (2023) describe such choices as a "largely counterfactual" case, and Cappelen et al. (2025) recode many of them as respondent confusion or enumerator error.

[10] In the early 20$^{\text{th}}$ century, people born in the US were more likely to express political discontent against immigration when the cultural difference between themselves and immigrants was higher (Tabellini, 2020). Having a more similar teacher leads to positive educational outcomes (Dee, 2004, 2005; Gershenson et al., 2022). Cultural distance affects trust and thus trade between countries (Guiso et al., 2009). More diversity—and thus less similarity—among residents in communities decreases the provision of public goods (Alesina et al., 1999), the formation of social capital (Alesina and La Ferrara, 2000), and trust (Alesina and La Ferrara, 2002; see also Alesina and La Ferrara, 2005 and Alesina and Giuliano, 2015 for reviews). Football players from a more similar cultural background are more likely to pass the ball to each other (Békés and Ottaviano, 2025).

# 3 Social discounting

The intuition behind social discounting is best captured by the following quote from Edgeworth (1881):

> ... between the frozen pole of egoism and the tropical expanse of utilitarianism, there [is a] position of one for whom in a calm moment his neighbour's happiness as compared with his own neither counts for nothing, nor yet 'counts for one,' but counts for a fraction. We must modify the utilitarian integral ... by multiplying each pleasure, except the pleasures of the agent himself, by a fraction—a factor doubtless diminishing with what may be called the social distance between the individual agent and those of whose pleasures he takes account.

It is this 'factor doubtless diminishing with ... social distance' that social discounting aims to model.

Consider a decision-maker $i$ who faces options characterised by two attributes, $x_j$ and $s_{ij}$. Attribute $x_j$ is an amount of money received by recipient $j$ while attribute $s_{ij}$ is the social distance between $i$ and $j$. We will proxy $s_{ij}$ via interpersonal similarity, how similar $i$ and $j$ are. Henceforth, $s_{ij} \in [0, 1]$ measures the dissimilarity between $i$ and $j$. If $s_{ij} = 0$, $i$ and $j$ are identical; if $s_{ij} = 1$, $i$ and $j$ are as dissimilar as possible.

According to social discounting, preferences over these options $(x_j, s_{ij})$ can be captured by the multiplicative utility function $U_i(x_j, s_{ij}) = D_i(s_{ij})\, u_i(x_j)$, where $D_i : [0, 1] \to \mathbb{R}_+$ is the social discount function and $u_i : \mathbb{R} \to \mathbb{R}$ is the utility of money. Since the decision-maker is identical to themselves, $s_{ii} = 0$, we follow Edgeworth (1881) and Sally (2001) by assuming that $D_i(0) = 1$: the decision-maker does not discount money they receive.

Note that $u_i$ only depends on $x_j$ and not on an hypothetical $x_i$. This means that the decision-maker does not compare what they get with what $j$ gets, which rules out, for example, inequality aversion. Similarly, $U_i$ only depends on $D_i$ and $u_i$ and not on $u_j$, meaning that the decision-maker judges $x_j$ using their own preferences and not $j$'s preferences. We will keep these assumptions throughout and thus write more succinctly $U(x, s) = D(s)\, u(x)$.

Our main objective is to identify the shape of the social discount function $D(s)$. Figure 1 represents different possible shapes. First of all, if $D(s)$ is flat, similarity plays no role. We thus have our first hypothesis:

**Hypothesis 1.** $D(s) = c$ for some constant $c \in [0, 1]$ and for all $s \in ]0, 1]$.

Hypothesis 1 implies that $D'(s) = 0$ for $s \in ]0, 1]$. Rejecting Hypothesis 1 would thus show that similarity matters. Social discounting itself corresponds to $D'(s) < 0$, while $D'(s) > 0$ would mean that the decision-maker places more weight on more dissimilar others, which we call 'social magnification'.

We will consider two special cases of Hypothesis 1. If $c = 0$, the decision-maker places no weight on others and is indifferent between any two options that benefit others. On the other hand, for any $c > 0$ the decision-maker always chooses the option that provides the largest amount of money, regardless of who receives it. We will return to these cases making them explicit when we introduce a functional form for $D(s)$.

# 4 Experimental design

The objective of our experiment is to identify the shape of the social discount function $D(s)$. We will present subjects with options that vary amounts of money $x$ and dissimilarity $s$ and observe their choices. In order to do so, we first need a way to measure interpersonal similarity and generate variation in it.

## 4.1 Measuring and varying interpersonal similarity

We measure and vary interpersonal similarity using a two-step process.

**Step 1: Gathering passive subjects.** We first surveyed members of the US population to create a set of passive subjects. We asked them 40 questions adapted from existing sources such as the American Census, the American Community Survey, the General Social Survey, and the World Value Survey. The questions are a mix of standard demographic questions—sex, age, location, race, ethnicity, religion, etc.—and questions about one's opinion on different topics—confidence in several institutions, trust in others, etc.[11]

**Step 2: Generating variation in similarity.** We then invited a second set of subjects and asked them to answer the same questions. As soon as each subject in Step 2 submitted their answers, we looked at the difference between their answers and the answers from all the passive subjects in Step 1.

In principle, we could compute these differences for any of the answers elicited from the survey. We chose the answers to the following subset of the questions: sex, date of

[11]The list of questions used in the survey appears in the Appendix of Beranek and Castillo (2024), which relied on the same survey.

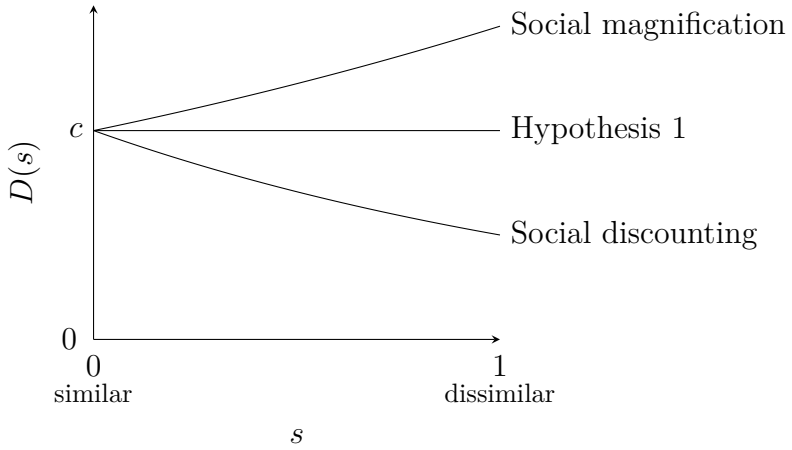



**Figure 1:** The social discount function $D(s)$ and Hypothesis 1, drawn for $c = 1$.

birth, zip code, race, ethnicity, religion, and the place where one grew up. Our choice was driven by the fact that we needed questions whose answers could be succinctly conveyed to subjects. We also wanted to use characteristics that have been found to be important and salient; these variables are all highlighted as such in the homophily literature (McPherson et al., 2001). Finally, we wanted to avoid loaded questions that might trigger an emotional response, such as support for the right to an abortion, the death penalty, or affirmative action.

For categorical variables, we computed the difference as 0 if the answers were the same and 1 if they were different. For example, if a subject in Step 2 answered 'white' to the question in which we asked their race, the difference for this variable would be 0 when compared to a passive subject in Step 1 who also answered 'white' and 1 when compared to a passive subject who answered something different. For characteristics elicited through a main question with a follow-up question, we computed a separate binary difference for the main question and for the follow-up question.[12]

For continuous variables, we normalised the absolute difference with the maximum difference observed for each subject in comparison to our sample of passive subjects to

[12]A match on the main question but a mismatch on the follow-up question, therefore, contributes a fractional difference for that characteristic. For example, for religion, the main question asks whether a subject belongs to a religious denomination and the follow-up asks which one. The difference is 0 if neither belongs to a religion or if both subjects belong to a religion and report the same denomination (for example, both 'Catholic'). The difference is 0.5 if both belong to a religion but report different denominations (for example, one 'Catholic' and one 'Protestant'). The difference is 1 if one subject belongs to a religion and the other does not. We apply the same procedure for ethnicity and the place where one grew up.

ensure that they are between 0 and 1. For example, for the question in which we asked for the zip code, we computed the geodesic distance between the two zip codes' centroids and normalised it by the maximum distance in our sample.

After having computed the difference for each of these variables, we took the unweighted average of these differences to create a *dissimilarity index*. The index measures how dissimilar the answers of a subject in Step 2 are compared to a given passive subject from Step 1. The index ranges between 0 and 1: an index of 0 describes a passive subject who has given exactly the same answers to every question while an index of 1 describes a passive subject who has given answers as different as possible.

Independently for each Step 2 subject, we computed the dissimilarity index towards all Step 1 passive subjects. Then, again independently for each Step 2 subject, we ordered all Step 1 passive subjects based on the values of the dissimilarity index from least to most dissimilar. Finally, to generate variation in similarity, we selected three subjects: the most similar, with the smallest dissimilarity index; the least similar, with the largest index; and the one in the middle, with an intermediate index.

In this paper, we refer to these selected subjects as 'matches', and use the shorthands 'similar match', 'intermediate match', and 'dissimilar match'. In the experiment itself, we randomly associated each match with one of the four suits from a deck of playing cards (♣ ♦ ♠ ♥) and referred to them as such to the Step 2 subjects using the card display shown in Figure 2. This way, we avoided using labels that are obviously ranked, such as $(1, 2, 3)$ or $(A, B, C)$, when referring to the matches during the experiment.

The variables shown on the card display are the ones we used to compute the dissimilarity index. However, we did not tell subjects about the index. If we had, subjects might mechanically have chosen options that benefit matches with low dissimilarity index. For similar reasons, we did not tell subjects how these matches were selected.

We revealed to subjects their three matches all at once in a randomised order as can be seen in Figure 2. Then, to make sure that subjects engaged with the specific characteristics of each of their matches, we presented the matches one by one in a randomised order and asked them to write in at least 25 characters the first things that came to mind when they read the card (Figure A4 in Appendix A shows an example).

## 4.2 Inclusion of Other in the Self task as a manipulation check

To make sure our manipulation was effective, we used the Continuous Inclusion of Other in the Self (IOS) scale (Beranek and Castillo, 2024). Compared to the standard IOS scale (Aron et al., 1992), the Continuous IOS scale provides a finer measurement and, as we

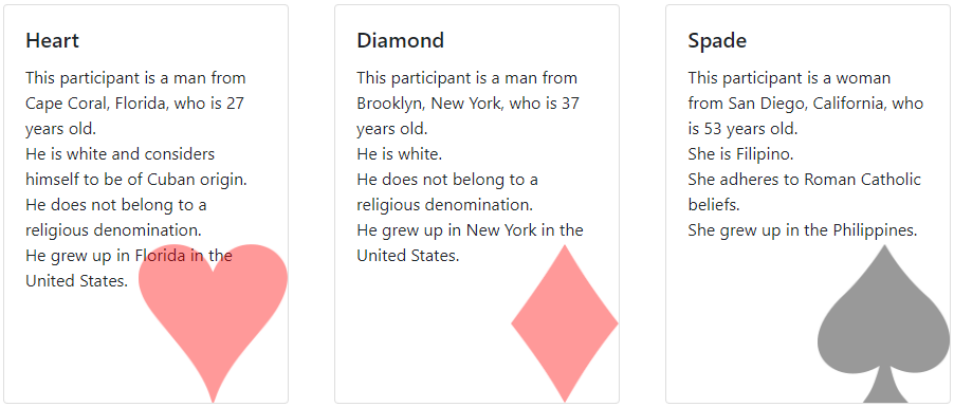



**Figure 2:** Card display used in the experiment.

show in Beranek and Castillo (2024), it solves a bias of the original IOS scale whereby subjects avoid selecting low levels of closeness.

Subjects evaluated each of their matches, one after the other and in a randomised order, using the Continuous IOS scale (Figure A5 in Appendix A shows an example). The measure obtained from the Continuous IOS scale is the degree of overlap between the two circles which represents the extent to which subjects feel connected to one of their matches. An overlap of 0 means that subjects do not feel connected, while an overlap of 1 means they feel very connected.

## 4.3 Choice task

Finally, we explain the main ingredient of our experiment: the choice task. In the choice task, subjects made a series of choices between two options. Each option features a specific amount of money going to one of their matches. Subjects could also report explicit indifference (Figure A6 in Appendix A provides an example).

We chose three amounts of money: \$1, \$2, and \$5. With three matches and three amounts of money, that makes 9 possible options. We generate all possible binary choices among these 9 options, remove choices where there is dominance in money (e.g., \$1 for the similar match vs \$2 for the similar match) but keep choices where there is dominance in interpersonal similarity (e.g., \$1 for the similar match vs \$1 for the dissimilar match). That leaves us with 27 choices.

To control for order effects within a choice, the position of each option—left or right—was randomised independently for each subject and for each choice. To control for order effects within the choice task—e.g., seeing options that benefit the similar match before seeing options that benefit the dissimilar match—subjects saw these 27 choices in an

order randomised independently for each subject. To control for order effects between tasks, the order of the IOS task and the choice task was also randomised. We show in Appendix B that there are no order effect between tasks, so we pool all the data in our analysis.

Our design has a number of additional features that allow us to isolate the effect of similarity. First, we control for selfish motives by having options that do not benefit the Step 2 subjects themselves. The subjects do not split a pie between themselves and their matches; instead, they face a series a binary dictator games with only other-other allocations. Second, we control for reciprocity by having options that benefit a pool of passive subjects different from those who are actively taking decisions. The Step 2 subjects knew that the targets of the options they were considering were passive subjects who had completed the survey long ago, and that the money they themselves received for their participation was not determined by the choice of another subject. Third, we control for reputation concerns by not revealing the characteristics of the Step 2 subjects to the passive subjects when making the payments. Otherwise, the Step 2 subjects might choose options based on what they think is expected of them—e.g., they might think that residents of New Jersey have to choose residents of New Jersey. We go one step further by not revealing to the passive subjects that an experiment even took place. The money the passive subjects received came directly from us without mention of any experiment, which we told the Step 2 subjects in our instructions. Therefore, it is not possible for the Step 2 subjects to choose a particular option to let a passive subject know that someone similar to them decided to send them money.

At the end of the experiment, we selected one of the 27 choices at random (also independently for each subject) and implemented the option selected. If a subject reported indifference in the selected choice, each option had a 50% chance of being implemented, as was explained in the instructions.

We took a number of steps to convince subjects that their choices had real consequences. Some of them are standard: we used an account that has been rated positively by previous subjects in other, unrelated experiments; our IRB approval was prominently featured; and we made clear in the instructions and the control questions our intention to implement the choices. We went one step further by posting on our website the receipts of all payments we made.[13] Subjects were invited to visit our website to check the receipts and convince themselves we actually pay. So that even the first subjects were convinced too, we first ran a few pilot sessions before the actual sessions and reported those payments. Finally,

[13]Visit `https://geoffreycastillo.com/mturk/` to see exactly what we showed the subjects.

we told subjects we would (and then we did) send them a notification once their payment was sent with a link to our website where they could literally check the receipts.

### 4.4 Implementation details

We conducted the experiment in December 2020 on Amazon MTurk. A total of 500 subjects participated in the experiment. We preregistered a data-cleaning protocol to remove from the sample subjects who submitted implausible or suspicious answers. After implementing this protocol, 355 subjects remain in the pool for analysis.[14] These subjects were all paid a flat fee of $3.00. We implemented one of the choices each of these subjects made and paid $3.24, on average, to their matches.

The experiment was programmed using oTree (Chen et al., 2016). The full instructions can be found in Appendix A.

## 5 Results

### 5.1 Similarity and Inclusion of Other in the Self scale

Before getting into our main analysis, we perform a series of simple checks. We first confirm that use of the dissimilarity index leads to the distinct matches we expected. Remember that for each subject we selected three matches based on their computed dissimilarity index towards the passive subjects. The left panel in Figure 3 reports a box plot of the dissimilarity index for each match. We see that, by construction, different matches have different levels of dissimilarity. The average dissimilarity index for similar, intermediate, and dissimilar matches is 0.056, 0.337 and 0.736, respectively. These values can be interpreted as the average degree of demographic mismatch, averaging categorical differences and normalised continuous distances, between subject and match on the variables shown in Figure 2—sex, age, geographic location, race, ethnicity, religion, and the place where one grew up.

Second, we check whether what we call similar, intermediate and dissimilar matches are perceived as such by subjects. To do that, the right panel in Figure 3 shows box plots of the Continuous IOS overlaps reported by subjects for each match. We see that

[14] We removed 87 obvious bots who, for example, copied-and pasted the instruction text or random text found online into any of their free text responses. We also removed 56 subjects who exhibited at least two instances of suspicious behaviour; for example, completing the entire survey in less than 3 minutes or having completed the experiment from the same IP address as another subject. All of these exclusion criteria were preregistered. We also exclude two subjects who managed to report a proportion of overlap greater than 1 in the Continuous IOS scale.

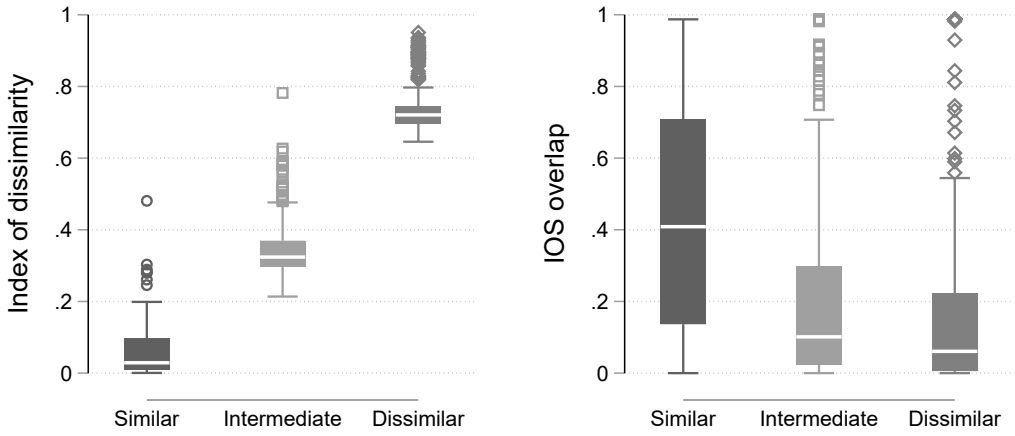



**Figure 3:** Box plots of the dissimilarity index and of the degree of overlap in the Continuous IOS scale for each match.

subjects report a greater degree of overlap with more similar matches, which indicates a greater connection in the context of the IOS scale. We confirm this using Page's (1963) trend test (Page's $L$ statistic = 4660, $p < 0.01$).

## 5.2 Choice data

Next, we look at the choices made by subjects in the choice task. In total, subjects made 27 choices, and we represent each of these choices in Figure 4. Each horizontal stacked bar represents one of the 27 choices. The proportion of subjects' choices are represented by the colour of the bars. The greater proportion of dark within a given bar, the more subjects favoured the more similar match in this choice.

Each set of stacked bars represents one of the nine possible combinations of money amounts. Within a given set, the top bar shows choices featuring the similar and the intermediate matches; the middle bar, the similar and the dissimilar matches; and the bottom bar, the intermediate and the dissimilar matches. The middle column in the figure shows choices in which both matches get the same amount of money. To the left of the middle column, more similar matches get more money; and conversely, to the right, the more dissimilar matches get more.

From this figure, we make four important observations. First, we see that subjects are not indifferent between all options. Instead, subjects respond to changes in the attributes. For example, increasing the amount of money provided to a given match increases the proportion of subjects who choose this match.

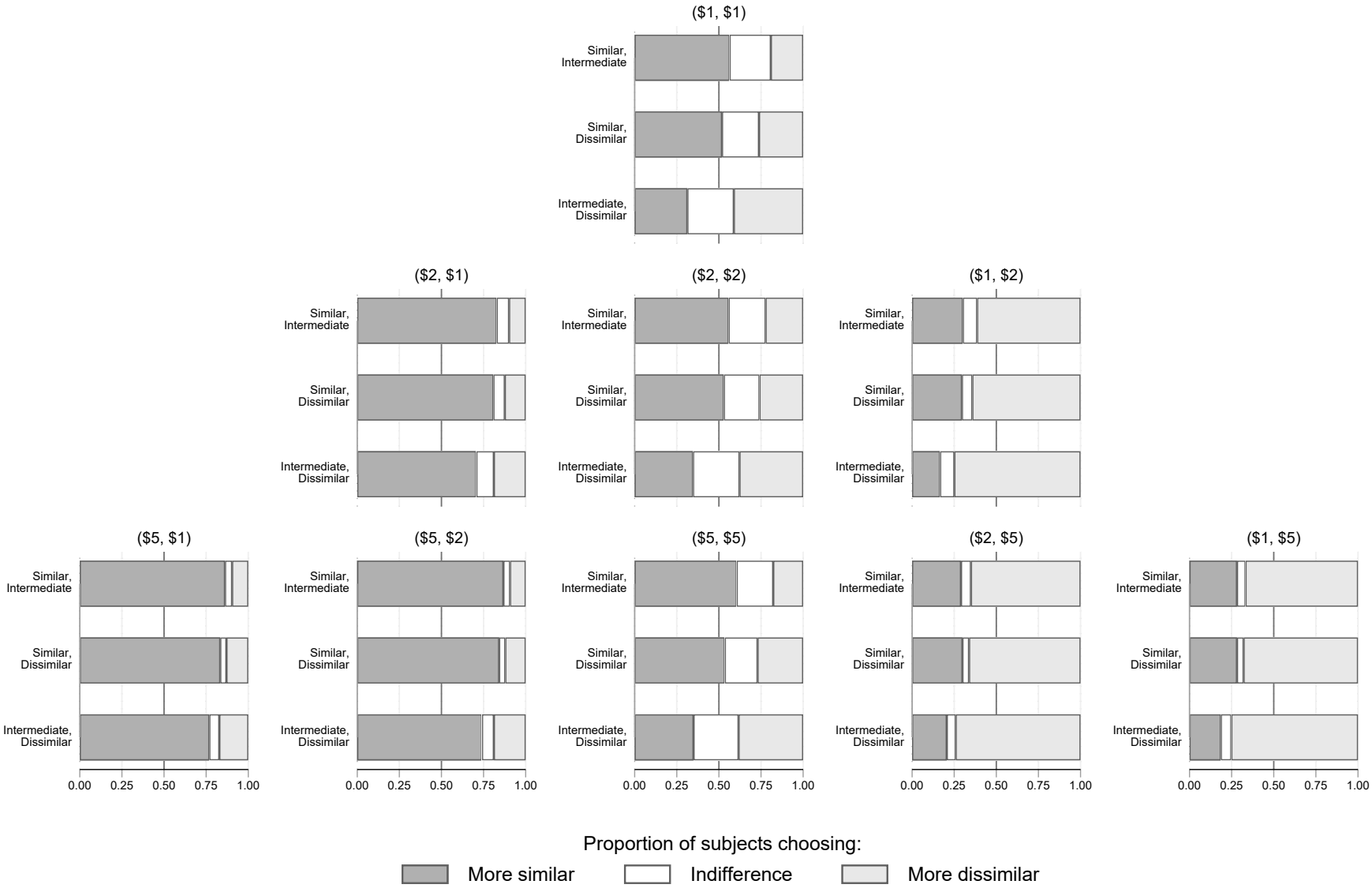



**Figure 4:** Stacked bar graphs showing proportion of choices over monetary amounts and matches.

Second, subjects often select more similar matches which is already evidence for social discounting. We observe in the right part of the figure that social discounting is sufficiently strong that there are subjects who choose the similar match when they could have provided two to five times as much to the intermediate or the dissimilar match.

However, the figure also suggests that there are subjects who prefer more dissimilar matches; the proportion of subjects who choose the more dissimilar match never goes to 0. For example, in the left part of the figure, where more similar matches get more money, there are still subjects who choose the option that favours the dissimilar match when they could have provided up to five times as much to the more similar match.

The last observation is that the preference for more similar matches is dampened when the most similar match is not involved. If we focus on the middle column, for instance, we see that most subjects prefer the similar match when choosing between the similar match and the intermediate match or between the similar match and the dissimilar match, but more prefer the dissimilar match when choosing between the intermediate match and the dissimilar match. This suggests that preferences are not monotonous and that more similar matches are not always preferred.

## 5.3 Nonparametric estimation

### 5.3.1 Functional forms and hypotheses

We then use this choice data to estimate the social discount function $D(s)$ nonparametrically. This nonparametric estimation allows us to let the data reveal the shape of $D(s)$ without making any specific assumption about its functional form.

Recall that each subject $i$ faces three matches $j$. Let $j = 1$ correspond to the similar match, $j = 2$, to the intermediate match, and $j = 3$, to the dissimilar match. We assume the utility function $U(x_j, \{d_{ij}\}_{j=1}^{3}) = d_{ij}x_j$ and estimate the three weights $d_{ij}$. Comparing the values of these weights as a function of the underlying $s_{ij}$ allows us to reveal the implicit shape of the social discount function $D(s)$.

Because of the assumption that $D(s) \geq 0$ (Section 5.4.1), we assume that these weights are weakly positive, $d_{ij} \geq 0 \, \forall \{i, j\}$. As can be seen, this utility function assumes that the utility function of money $u(x)$ is linear. We will relax this assumption later when we will move on to parametric estimations.

### 5.3.2 Stochastic assumption and likelihood

To take the model to the data, we make the following stochastic assumptions. In the experiment, subject $i$ faces choice tasks $t \in \{1, \ldots, 27\}$. In each choice task (described in Section 4.3), subjects choose between a left and a right option, which we denote by $\omega_t^L$ and $\omega_t^R$. Subjects can also report indifference, which we write as $\iota_t$. Denote the utility difference between $\omega_t^L$ and $\omega_t^R$ by

$$\Delta U_t\Big(\{d_{ij}\}_{j=1}^3\Big) = U\Big(\omega_t^L; \{d_{ij}\}_{j=1}^3\Big) - U\Big(\omega_t^R; \{d_{ij}\}_{j=1}^3\Big).$$

We then use the Fechner model (Hey and Orme, 1994) and assume that subjects make stochastic errors $\epsilon \sim \mathcal{N}(0,1)$ when judging the utility difference such that $\omega_t^L$ is chosen when $\Delta U_t\Big(\{d_{ij}\}_{j=1}^3\Big) + \epsilon > 0$.

Based on this stochastic model, the probability of observing subject $i$ in choice $t$ choosing $\omega_t^L$ is

$$\Pr\Big(\Delta U_t\Big(\{d_{ij}\}_{j=1}^3\Big) + \epsilon > 0\Big) = \Phi\Big(\Delta U_t\Big(\{d_{ij}\}_{j=1}^3\Big)\Big),$$

where $\Phi$ is the cumulative distribution function of the standard normal distribution. The probability of choosing $\omega_t^R$ is similarly derived. Denoting the choice made by a subject by $c_t$ and ignoring for now the choices where subjects reported indifference, the log-likelihood of subject $i$ over their $t$ choices is

$$\begin{aligned}\ln L_i = \sum_{t=1}^{27} \Bigg[ &\left( \ln \Phi\Big(\Delta U_t\Big(\{d_{ij}\}_{j=1}^3\Big)\Big) \;\middle|\; c_t = \omega_t^L \right) \\ &+ \left( \ln \Phi\Big(-\Delta U_t\Big(\{d_{ij}\}_{j=1}^3\Big)\Big) \;\middle|\; c_t = \omega_t^R \right) \Bigg].\end{aligned}$$

If they reported indifference, subjects knew that each option had a 50% chance of being implemented. To take this into account, we follow Andersen et al. (2008), based on the fractional response approach of Papke and Wooldridge (1996), and augment the

log-likelihood by giving each option an equal weight:

$$\ln L_i = \sum_{t=1}^{27} \Bigg[ \left( \ln \Phi\Big( \Delta U_t\big(\{d_{ij}\}_{j=1}^3\big) \Big) \;\Big|\; c_t = \omega_t^L \right) + \left( \ln \Phi\Big( -\Delta U_t\big(\{d_{ij}\}_{j=1}^3\big) \Big) \;\Big|\; c_t = \omega_t^R \right) + \left( \tfrac{1}{2} \ln \Phi\Big( \Delta U_t\big(\{d_{ij}\}_{j=1}^3\big) \Big) + \tfrac{1}{2} \ln \Phi\Big( -\Delta U_t\big(\{d_{ij}\}_{j=1}^3\big) \Big) \;\Big|\; c_t = \iota_t \right) \Bigg].$$

We estimate $\{d_{ij}\}_{j=1}^3$ via maximum likelihood independently for each subject.[15] For ease of comparison between subjects, we normalise the $d_{ij}$ such that they are always at most 1, i.e., $d_{ij} / \max_j d_{ij}$ for each subject $i$.

### 5.3.3 Estimation results

Once we have these estimates, for each subject $i$ we look at the ordering of their $\{d_{ij}\}_{j=1}^3$. This allows us to reveal the type of subject $i$. For example, if $d_{i1} > d_{i2} > d_{i3}$, then subject $i$ is a monotone social discounter who discounts more as similarity decreases.

We first categorise subjects based solely on the ordering between their similar and dissimilar matches, irrespective of the position of the intermediate match. By looking at the two extremes of the similarity scale, we essentially replicate what most of the previous literature has done. We call subjects with $d_{i1} > d_{i3}$ 'social discounters' and subjects with $d_{i3} > d_{i1}$ 'social magnifiers'. We find that about 60% of subjects are social discounters and about 29% are social magnifiers.

We next consider the intermediate match $d_{i2}$ and categorise subjects based on the

[15] We use **R** (R Core Team, 2023) with the `maxLik` package (Henningsen and Toomet, 2011). We rely on the BFGS algorithm with numerical derivatives. For each subject, we run the estimation with 80 randomly selected starting values. We keep the estimates with the maximum likelihood, for which all standard errors can be recovered, and whose Hessian is negative definite. When comparing the estimated weights, we treat two weights as equal when, after the normalisation, they differ by less than $10^{-4}$, which is the numerical precision of the estimates.

We can recover valid estimates from the maximum likelihood estimation for 314 out of 355 subjects. Among them, 6 are indifferent in all their choices and thus have all their $d_{ij}$ estimated at zero: payoffs to others do not enter their utility. We report them as a separate category. The remaining 41 subjects make choices that some set of weights $d_{ij}$ explains perfectly. As with perfect separation in a logit, scaling these weights up always increases the likelihood, so it has no maximum. For them, we recover the ordering of the $d_{ij}$ by revealed preference. Each choice bounds the ratio of two weights: for example, choosing \$1 for the similar match over \$2 for the dissimilar match reveals $d_{i1} > 2d_{i3}$. Combined, these bounds determine the full ordering for all 41 subjects. These 41 subjects are not shown in Figure 5 since they do not have $d_{ij}$ estimates, but we include them when reporting the proportions of subjects in each ordering.

implicit shape of their social discount function. We find that about 35% of subjects would have a U-shaped social discount function, placing less weight on the intermediate match than on any other match. About 35% of subjects would have a monotone social discount function, either increasing or decreasing. Finally, about 14% of subjects would have an inverse U-shaped social discount function, placing more weight on the intermediate match than on any other match.

These two categorisations combine to generate different orderings and thus different groups of subjects. We focus on the six most common orderings, which together account for about 88% of subjects, and plot the $d_{ij}$ as a function of the dissimilarity index $s_{ij}$ in Figure 5 for each of them. Each subject appears three times, once for each match $j$, in each of these graphs. To get a sense of the shape of a social discount function $D(s)$ that would fit the data, we also add a nonparametric regression line. In Appendix C we show all subjects and all orderings in table form.

We see that the most common ordering is $d_{i1} > d_{i2} > d_{i3}$, which corresponds to subjects who are monotone social discounters. The second most ordering, $d_{i1} > d_{i3} > d_{i2}$, corresponds to subjects with a U-shape social discount function. They are broadly discounters in the sense that the similar match is the least discounted, but they discount the intermediate match more than the dissimilar match.

We then see emerge orderings that reveal subjects who put more weight on more dissimilar matches. For example, subjects with ordering $d_{i3} > d_{i1} > d_{i2}$ put more weight on the dissimilar match than on the other matches, but then exhibit social discounting between the similar match and the intermediate match—their social discount function is also U-shaped. Subjects with ordering $d_{i3} > d_{i2} > d_{i1}$ are the mirror of monotone social discounters, and can be referred to as monotone social magnifiers.

## 5.4 Parametric estimation

We now move to the core of our analysis, the maximum likelihood estimation of the structural parameters of social discounting. We have just observed two important features of our data: the social discount function is not necessarily monotone, and there is considerable heterogeneity among subjects in the shape of the social discount function. In response to these features, we use flexible functional forms and we adopt a random coefficient framework.

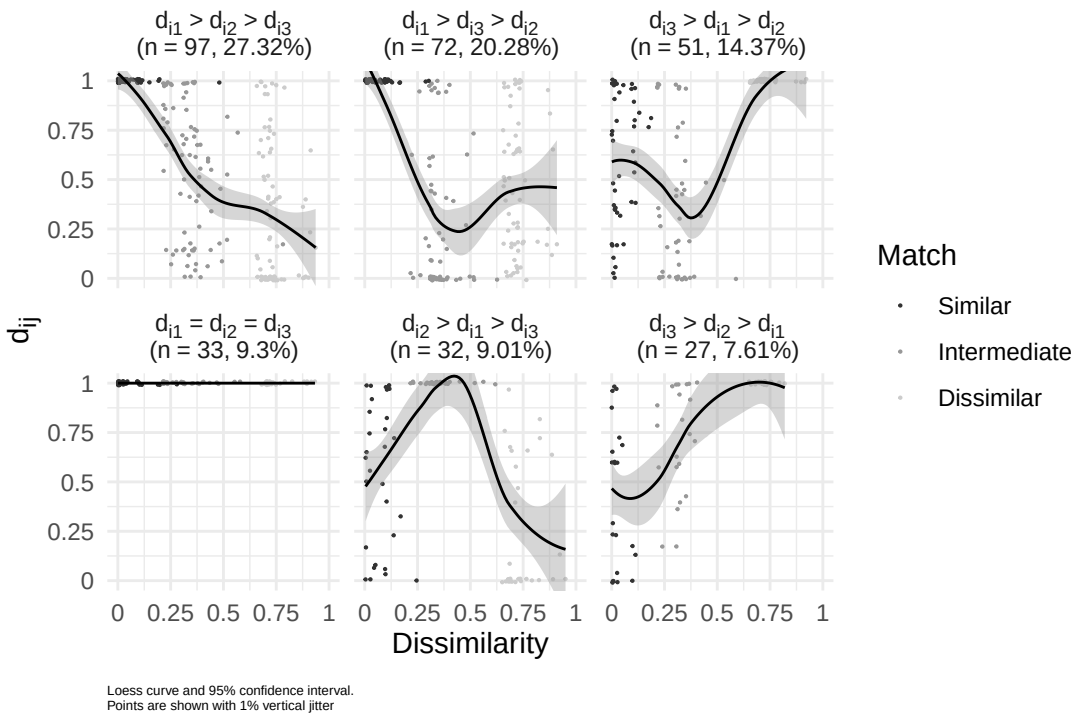



**Figure 5:** Scatterplot of dissimilarity index and $d_{ij}$ estimated, separately for each of the most common orderings.

### 5.4.1 Functional forms and hypotheses

To control for the curvature of $u(x)$, we choose the exponential utility function

$$u(x, \alpha_i) = \begin{cases} 1 - \exp(-\alpha_i x) \text{ if } \alpha_i > 0, \\ x \text{ if } \alpha_i = 0, \\ \exp(-\alpha_i x) - 1 \text{ if } \alpha_i < 0, \end{cases} \tag{1}$$

where $\alpha_i$ captures the curvature. $\alpha_i > 0$ gives rise to a concave utility function, $\alpha_i < 0$ to a convex utility function, and $\alpha_i = 0$ to a linear utility function.

For the social discount function $D(s)$, we will consider two functional forms. The first one is the exponential discount function $D(s) = \delta^s$ where $\delta \geq 0$ is the social discount factor. This particular form mirrors the classic exponential discount function of the Discounted Utility model in intertemporal choice (Koopmans, 1960; Samuelson, 1937). It is monotone, either strictly decreasing for $\delta \in ]0, 1[$ or strictly increasing for $\delta > 1$.

However, as we have seen in the previous section, we do not necessarily expect the social discount function to be monotone. Therefore, we also consider a more flexible

functional form that includes a quadratic term: $D(s) = \delta^{s+\kappa s^2}$, where $\kappa \in \mathbb{R}$ is an additional curvature parameter. $\kappa < 0$ increases the curvature of the social discount function, while $\kappa > 0$ decreases it. When $\kappa < -1/2$, the stationary point of the curve is between 0 and 1. Therefore, when $\delta \in ]0, 1[$ and $\kappa < -1/2$, the social discount function is U-shaped. When $\delta > 1$ and $\kappa < -1/2$, the social discount function is inverse U-shaped. We represent these cases in Figure 6.

With these functional forms, we can highlight two special cases of Hypothesis 1:

**Hypothesis 2.** $\delta = 0$,

**Hypothesis 3.** $\delta = 1$.

If $\delta = 0$, then subjects do not care about payoffs given to others at all. If $\delta = 1$, then subjects do not discount payoffs given to others based on similarity. If we reject both Hypotheses 2 and 3, we reject in full Hypothesis 1 that the social discount function $D(s)$ is flat.[16]

Nonetheless, as a benchmark we also estimate a model in which we force the social discount function to be flat, i.e., $D(s) = 1$. This benchmark allows us to establish how much better the exponential and quadratic-exponential social discount functions fit the data compared to a flat social discount function. Another possible benchmark would be that subjects do not care about payoffs given to others at all. However, if that was the case, we would have observed many subjects reporting indifference in all their choices which the choice data rules out: only 6 subjects out of 355 do so.

### 5.4.2 Stochastic assumptions and likelihood

We make similar assumptions as in the nonparametric estimation regarding the stochasticity of choices. We continue using the Fechner model so the contribution of choice $t$ to the likelihood is derived similarly. Indifference is again treated as a fractional response with each option receiving an equal weight. This time, we assume that $\epsilon \sim \mathcal{N}(0, \sigma^2)$ and so also estimate the standard deviation $\sigma$.

To account for heterogeneity, we follow a random coefficient approach: we assume that the parameters are distributed according to some distribution and estimate the

[16] $D(s) = \delta^s$ is flat only at $\delta = 0$, which gives $c = 0$, and at $\delta = 1$, which gives $c = 1$. The intermediate cases $c \in ]0, 1[$ share their observable implication with $c = 1$: in our design every option pays one of the matches and none pays the subject, so a flat $D(s) = c$ scales every utility difference by the same $c$. Choices then respond to the amounts of money and not to $s$. Rejecting $\delta = 1$ thus rejects a flat social discount function at any strictly positive level, and rejecting $\delta = 0$ rules out the one remaining case.

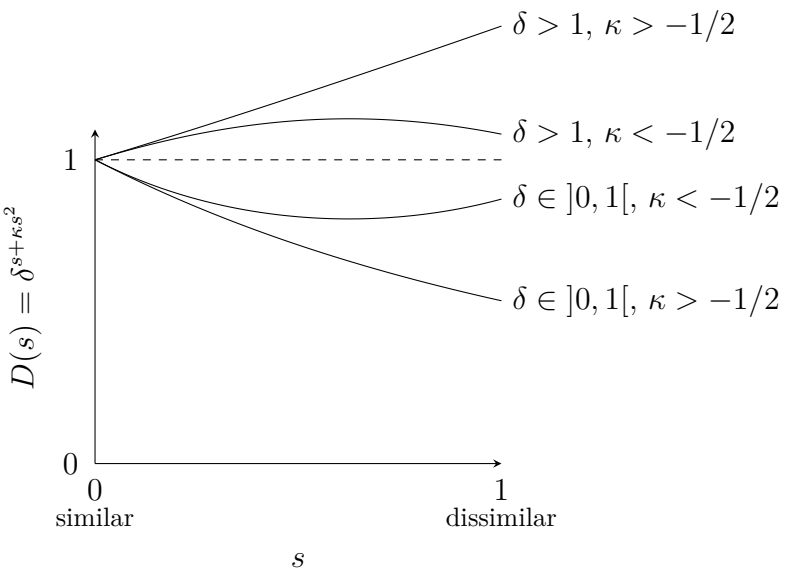



**Figure 6:** The quadratic-exponential social discount function.

parameters of these distributions. Compared to other approaches, such as estimating the parameters independently for each subject or estimating a single value for each parameter for the whole sample, this approach allows us to both capture heterogeneity while keeping a small degree of freedom (see Conte et al., 2011, for a discussion of this approach compared to other approaches).

Consequently, we assume that $\alpha$ is normally distributed, $\alpha \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$. Since the exponential social discount function imposes the requirement $\delta \geq 0$, we assume that $\delta$ is log-normally distributed, $\ln \delta \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2)$. Furthermore, for the quadratic-exponential social discount function, we assume that the curvature parameter $\kappa$ is normally distributed, $\kappa \sim \mathcal{N}(\mu_\kappa, \sigma_\kappa^2)$. We assume that these parameters are distributed independently to keep the number of estimated parameters small.

Denote by $\theta$ the parameters considered, and by $\tilde{\theta}$ the vector of moments of their distributions to be estimated. For example, we have $\theta = (\alpha, \delta)$ and $\tilde{\theta} = (\mu_\alpha, \sigma_\alpha, \mu_\delta, \sigma_\delta)$ for the exponential social discount function. The contribution to the likelihood of the choices $t$ of subject $i$ is then

$$L_i = \int_{\mathbb{R}} \cdots \int \prod_{t=1}^{27} L\left(c_t; \theta_i, \sigma\right) f(\theta; \tilde{\theta}) \, \mathrm{d}\theta$$

where $f$ is the joint density of the parameters. Finally, the sample log-likelihood is

$\ln L = \sum_{i=1}^{I} \ln L_i$. We estimate $(\tilde{\theta}, \sigma)$ via the method of maximum simulated likelihood (in the manner of Conte et al., 2011, and von Gaudecker et al., 2011).[17]

### 5.4.3 Estimation results

Table 1 reports the estimation results. The first column shows the results for the benchmark model of flat discounting. The second column shows the results of the model with the exponential social discount function, while the third column shows the results with the quadratic-exponential social discount function.

Since $\delta$ is log-normally distributed, it is hard to infer the shape of its distribution from these estimates alone. Therefore, we also plot the estimated distribution of the parameters in Figure 7. Table 2 further reports the mean, median and mode of the distribution of $\delta$ and the results of hypothesis tests of these moments against 1, which corresponds to no discounting.[18]

There are two main takeaways from Figure 7 and Table 2. First, no matter the social discount function used, most of the distribution of $\delta$ is between 0 and 1. We find no empirical support for $\delta = 0$ (Hypothesis 2): the estimated moments are far from zero, and as Figure 7 illustrates, the bulk of the estimated distribution lies strictly positive. This aligns with our non-parametric results showing that subjects place positive weight on

[17] We use Halton sequences of length 100 per subject. To avoid local maxima, we follow the procedure of Liu and Mahmassani (2000) highlighted by Train (2009): we use a genetic algorithm (**R** package `GA` package, Scrucca, 2013, 2017) with wide search bounds, a large population size (4000) and a low number of iterations (5) to find 20 starting values which we feed to the BFGS algorithm for a standard maximum likelihood estimation. We keep the estimates with the maximum likelihood, for which all standard errors can be recovered, and whose Hessian is negative definite.

[18] The moments are computed as mean $= \exp\left(\mu + \frac{\sigma^2}{2}\right)$, median $= \exp(\mu)$, and mode $= \exp\left(\mu - \sigma^2\right)$.

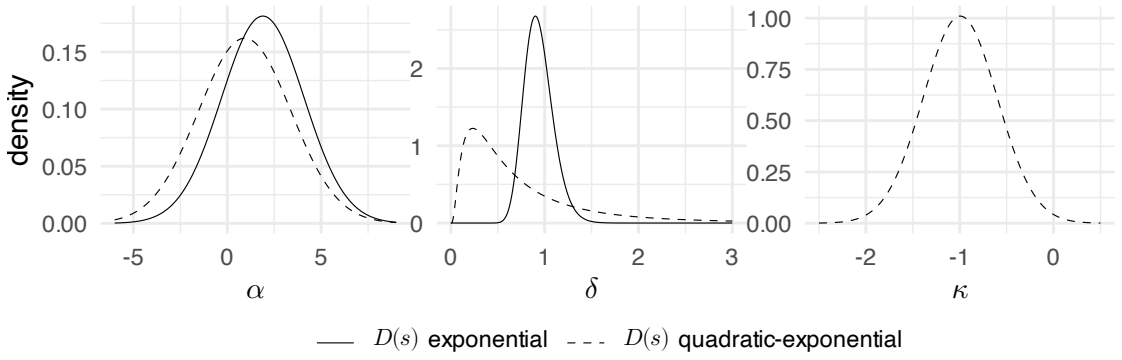



**Figure 7:** Plots of the distributions estimated in Table 1 of $\alpha$ (utility of money curvature), $\delta$ (social discount factor), and $\kappa$ (quadratic parameter).

**Table 1:** Maximum simulated likelihood estimates.

| $D(s)$ | No discounting | Exponential | Quadratic-exponential |
|---|---|---|---|
| $\mu_\alpha$ | $1.02^{***}$ | $1.89^{***}$ | $0.93^{***}$ |
| | (0.26) | (0.16) | (0.07) |
| $\sigma_\alpha$ | $3.58^{***}$ | $2.20^{***}$ | $2.46^{***}$ |
| | (0.82) | (0.18) | (0.16) |
| $\mu_\delta$ | | $-0.08^{***}$ | $-0.61^{***}$ |
| | | (0.01) | (0.07) |
| $\sigma_\delta$ | | $0.16^{***}$ | $0.92^{***}$ |
| | | (0.02) | (0.10) |
| $\mu_\kappa$ | | | $-0.99^{***}$ |
| | | | (0.01) |
| $\sigma_\kappa$ | | | $0.39^{***}$ |
| | | | (0.01) |
| $\sigma$ | $0.24^{***}$ | $0.10^{***}$ | $0.10^{***}$ |
| | (0.03) | (0.01) | (0.01) |
| Log Likelihood | $-5011.41$ | $-4539.03$ | $-4068.09$ |

*Notes.* Robust standard errors in parentheses.
$\alpha$ and $\kappa$ normally distributed, $\delta$ log-normally distributed.
$u(x)$ exponential as defined in (1).
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

**Table 2:** Moments of the distribution of $\delta$ and hypothesis tests.

| $D(s)$ | Moment of $\delta$ | | Test against 1 | |
|---|---|---|---|---|
| | | | $\chi^2$ | $p$ |
| Exponential | mean | 0.94 | 121.46 | $< 0.001$ |
| Exponential | median | 0.93 | 77.41 | $< 0.001$ |
| Exponential | mode | 0.90 | 50.47 | $< 0.001$ |
| Quadratic-exponential | mean | 0.82 | 80.58 | $< 0.001$ |
| Quadratic-exponential | median | 0.54 | 166.77 | $< 0.001$ |
| Quadratic-exponential | mode | 0.23 | 178.75 | $< 0.001$ |

*Notes.* Tests are two-sided Wald tests based on robust standard errors.

other recipients. We also reject $\delta = 1$ (Hypothesis 3) for all moments of the distribution of $\delta$ and for both functional forms of $D(s)$.[19] All moments are below 1, while the distribution of $\delta$ in the quadratic-exponential case is more right-skewed as can be seen from Figure 7. As can be seen in Table 2, we find a median value of $\delta$ of about 0.93 for the exponential social discount function and of 0.54 for the quadratic-exponential social discount function. This finding confirms that most subjects are social discounters.

The use of a structural model allows us to define and compute a self equivalent and a dissimilarity premium, akin to the certainty equivalent and the risk premium in the context of risk preferences. The self equivalent $SE$ is the amount of money that makes a subject indifferent between receiving this amount and seeing another person at dissimilarity $s$ receive \$$x$: $SE(x, s) = u^{-1}\Big(D(s)u(x)\Big)$. The dissimilarity premium is then the difference between these two amounts: $x - SE(x, s)$. It captures how much money a subject is willing to forgo in order to redirect a reduced amount to themselves instead of seeing the whole amount go to another person at dissimilarity $s$. It is positive when $D(s) < 1$, as is the case for social discounters, and negative when $D(s) > 1$, as is the case for social magnifiers.

To compute it, for each model we start by recovering the posterior means of the parameters for each subject conditional on their 27 choices (Conte et al., 2011; Train, 2009). We then compute the dissimilarity premium for each subject by plugging these posterior means into the formula above, using $x = 1$ and $s = 0.7$ (which is approximately the average dissimilarity index of the dissimilar match, as we saw in Section 5.1). We use the same posterior means to classify each subject as a social discounter or a social magnifier: subject $i$ is a social magnifier if $D(s_{i3}) > D(s_{i1})$, that is, if they place more weight on the dissimilar match than on the similar match. For social discounters, we obtain a mean dissimilarity premium of about \$0.24 (95% confidence interval: \$0.21–\$0.26) for the exponential social discount function and of about \$0.28 (95% confidence interval: \$0.25–\$0.31) for the quadratic-exponential social discount function. For social magnifiers, we obtain a mean dissimilarity premium of about −\$0.11 (95% confidence interval: −\$0.14 to −\$0.07) for the exponential social discount function and of about −\$0.06 (95% confidence interval: −\$0.11 to −\$0.03) for the quadratic-exponential social discount function.[20] These calculations mean that, on average, social discounters would

[19] Another way to test the null hypothesis $\delta = 1$ is to test the null hypotheses $\mu_\delta = 0$ and $\sigma_\delta = 0$ for the log-normal distribution of $\delta$. These tests are readily available in Table 1 and all subsequent tables of the same type. We always reject such nulls.

[20] The confidence intervals are obtained by bootstrapping subjects within each type: we draw 10,000 samples of subjects with replacement and recompute the mean dissimilarity premium in each sample. The means exclude the subjects whose dissimilarity premium is not defined for some $s$ between 0

be willing to forgo any amount up to \$0.24–\$0.28 of the \$1 to have it redirected to themselves—and thus would be willing to accept as little as \$0.72–\$0.76—rather than letting the dissimilar match have the \$1. Social magnifiers, on the other hand, would rather let the dissimilar match have the \$1 than receive up to \$1.06–\$1.11 themselves.

The second takeaway is that some of the distribution of $\delta$ is above 1, both for the exponential and the quadratic-exponential social discount functions. $\delta > 1$ corresponds to subjects who are social magnifiers: they do not discount but magnify money received by others as a function of similarity. We find that about 32% of the distribution of $\delta$ is above 1 in the case of the exponential social discount function, and about 37% in the case of the quadratic-exponential social discount function.[21] These numbers are similar to the proportion of subjects that we found to be social magnifiers in the nonparametric estimation (29%).

### 5.4.4 Model comparison

Next we assess the fit of the models reported in Tables 1 and 2. Since the models are nested, we can readily use the tests reported in these tables to compare the models. We see that we reject the null hypothesis that $\delta = 1$ for all moments of $\delta$, which favours exponential discounting over no discounting. Further, we reject the null hypothesis that $\mu_\kappa = 0$, which favours quadratic-exponential discounting over exponential discounting, confirming the

and 0.9: 33 of the 85 social magnifiers for the exponential social discount function, and 37 of the 101 social magnifiers and 12 of the 254 social discounters for the quadratic-exponential social discount function. For these subjects, $D(s)u(1)$ exceeds the upper bound of their concave utility function, so that no amount of money received by themselves is worth as much as \$1 received by the other person. Most of them are strong social magnifiers, while the social discounters among them have a non-monotone social discount function that rises above 1 elsewhere between 0 and 0.9. Because the strongest social magnifiers are excluded, the mean dissimilarity premium of social magnifiers is closer to zero than their median, which is −\$0.16 for the exponential social discount function and −\$0.08 for the quadratic-exponential social discount function. Pooling both types, the mean dissimilarity premium is about \$0.18 for the exponential social discount function and \$0.21 for the quadratic-exponential social discount function.

[21] In the quadratic-exponential case, $\delta > 1$ is not sufficient for being a social magnifier because the curvature introduced by $\kappa$ can be strong enough to overturn the ordering it implies: if $\delta > 1$ but $\kappa$ is sufficiently negative, $D(s)$ is inverse U-shaped and can turn down before $s_{i3}$, so that $D(s_{i3}) < D(s_{i1})$ and the subject discounts despite $\delta > 1$; conversely, if $\delta < 1$ but $\kappa$ is sufficiently negative, $D(s)$ is U-shaped and can rise back above $D(s_{i1})$ by $s_{i3}$, so that the subject magnifies despite $\delta < 1$. Solving $D(s_{i3}) > D(s_{i1})$ gives the condition $\ln \delta \left[1 + \kappa(s_{i1} + s_{i3})\right] > 0$, meaning that subject $i$ is a social magnifier if either $\delta > 1$ and $\kappa > -1/(s_{i1} + s_{i3})$, or $\delta < 1$ and $\kappa < -1/(s_{i1} + s_{i3})$. Since $\delta$ and $\kappa$ are estimated as independent random coefficients, the probability that a random draw from the estimated distributions is a social magnifier, for a given pair $(s_{i1}, s_{i3})$, is $\Pr(\delta > 1)\ \Pr\big(\kappa > -1/(s_{i1} + s_{i3})\big) + \Pr(\delta < 1)\ \Pr\big(\kappa < -1/(s_{i1} + s_{i3})\big)$, where each probability is computed from the estimated distributions of $\delta$ and $\kappa$. To compute the proportion of magnifiers we report, we evaluate this probability at each subject's own $(s_{i1}, s_{i3})$ and average across subjects.

nonparametric estimation that the social discount function is not necessarily monotone.

Another way of assessing fit is through likelihood-ratio tests, which we report in Table 3. As we can see, and confirming the previous tests, exponential discounting fits better than no discounting, and quadratic-exponential discounting fits better than exponential discounting.[22]

To give a better sense of how well the models fit, Table 3 also reports hit rates. To compute them, we use the posterior estimates mentioned above and compare the choice made by each subject in each choice task with the choice predicted by the model. Finally, we compute the hit rate as the proportion of choices predicted correctly by the model. Table 3 reports the average hit rate for each model. We see that, while the no discounting model predicts 58% of choices correctly, exponential discounting predicts 73% of choices correctly, an increase of 15 percentage points. Adding the quadratic term to the social discount function further increases the hit rate to 78%, an increase of 5 percentage points.

# 6 Hyperbolic social discounting

The evidence presented so far rests on two interwoven assumptions. On the experimental side, subjects never make trade-offs between money for themselves and money for others. As a result, we cannot identify any 'self bias' that subjects might have, i.e., the possibility that they place a larger weight on themselves than on others. This bias mirrors the classical 'present bias' in the intertemporal choice literature in which subjects place a larger weight on the present than on the future. On the modelling side, the social discount function is either exponential or quadratic-exponential. This excludes other popular functional forms in which the discount rate is not constant but exhibits a discontinuity such as quasi-hyperbolic discounting.

We designed an extension of our original experiment in which we include the self as a possible recipient to address both assumptions. The data from this extension experiment allows us to identify self bias and to estimate a quasi-hyperbolic social discount function.

[22] One model nests another when the standard deviation of one of the parameters is set to zero, which is a boundary of the parameter space. In that case, the correct asymptotic reference distribution is instead a mixture of $\chi^2$ distributions with lower degrees of freedom (Self and Liang, 1987). The $\chi^2$ distribution we report stochastically dominates the correct asymptotic reference distribution, so our $p$-values are conservative. Given the size of the likelihood-ratio statistics, this does not affect the ranking of the models. The same reasoning applies to Table 6.

**Table 3:** Comparison of the fit of the models estimated in Table 1.

| $D(s)$ | Log-likelihood | Likelihood-ratio test | | Hit rate |
|---|---|---|---|---|
| | | $\chi^2$ | $p$ | |
| No discounting | −5011.41 | | | 0.58 |
| Exponential | −4539.03 | 944.76 | $< 0.001$ | 0.73 |
| Quadratic-exponential | −4068.09 | 941.88 | $< 0.001$ | 0.78 |

*Notes.* The likelihood-ratio test compares the model in one row to the model in the row above it. Hit rate is the proportion of choices correctly predicted by the model.

## 6.1 Design of the extension experiment

From a design perspective, the first difference between our original experiment and the extension is that subjects participate to two sessions instead of one in the extension. The first session is identical in all respects to the original experiment: we select from the ordered passive subjects the most similar passive subject, the least similar passive subject, and the one in the middle who is 50% of the way to the least similar passive subject.

The second difference is that, in the second session, we include the subjects themselves as possible recipients. This is what allows us to identify self bias. As a result, in some choices, subjects now choose between money for themselves or money for one of their matches. In order to not present subjects with the exact same matches in both sessions, we instead select the second-most similar passive subject, the passive subject 25% of the way to the least similar passive subject, and the passive subject 75% of the way. As before, we create all possible options and all possible choices between these options which doubles the number of choices in the second session (54 in the second session compared to 27 in the first session) meaning that overall we have 81 choices per subject.

This extension was conducted in March 2021.[23] In total 121 subjects participated to both sessions. After implementing the same preregistered data-cleaning protocol as for our original experiment, we are left with 109 subjects whose decisions we analyse.[24] As before, we implemented one randomly selected choice for each subject and paid $3.25, on average, to whoever they selected in that choice—either one of their matches or themselves. Separately, subjects themselves were paid a flat fee of $3.00 for each session

[23]We used the same eligibility criteria as in the original experiment and also excluded workers who had taken part in it.

[24]In total, 215 subjects participated in at least one of the two sessions. When we implement the protocol on the 215 subjects, we remove 36 obvious bots and 11 subjects who exhibited at least two instances of suspicious behaviour, leaving 168 subjects. Among them, 109 (64.9%) completed both sessions.

they completed.

## 6.2 Estimation and results

Here we focus on the parametric estimation of the social discount function and consider a quasi-hyperbolic form to capture self bias.[25] The parametric estimation follows closely the estimation procedure we used for our original experiment (Section 5.4). To capture self bias, we use the quasi-hyperbolic social discount function

$$D(s) = \begin{cases} 1 \text{ if } s = 0, \\ \beta\, \delta^s \text{ if } s > 0 \end{cases} \tag{2}$$

where $s = 0$ represents the self while $s > 0$ represents others. This function mirrors the quasi-hyperbolic discount function used in intertemporal choice (Laibson, 1997). The term $\beta$ captures self bias and represents the premium one places on oneself. If $\beta = 1$ we go back to the social discount function we have already used. If instead $0 < \beta < 1$, there is a discontinuity at $s = 0$, which means the self is given a premium. In terms of Figure 1, this would correspond to $c < 1$.

Due to the requirement that $\beta > 0$ and $D(s) > 0$, we assume that $\beta$ is lognormally distributed, $\ln \beta \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2)$, and we estimate $(\mu_\beta, \sigma_\beta)$ in addition to the parameters we previously estimated. All other aspects of the estimation remain the same as before. We again estimate a model without discounting as well as models that include a quadratic term for the dissimilarity index.

Table 4 reports the estimation results, and Figure 8, plots of the estimated distributions. When we estimate the distribution of $\delta$ using an exponential social discount function without including the $\beta$ term representing self, the distribution of $\delta$ is heavily right-skewed with a mass close to 0. When we allow for self bias and consider the quasi-hyperbolic social discount function, the skewness of the distribution decreases. Still, Table 5 shows that the moments of the distribution of $\delta$ are significantly different from 1. This means that we still find evidence of social discounting even when we control for self bias.

Comparing the distribution of $\delta$ shown in Figure 8 with the distribution of $\delta$ we

[25] In Appendix D we report the choice data (as we did in Section 5.2 for the original experiment) and the nonparametric estimation (as we did in Section 5.3 for the original experiment) applied to this extension experiment. The results are in line with the results we obtained in our original experiment. For example, looking at the weights $d_{ij}$ placed on the matches, we find similar orderings in similar proportions. In particular, we find a proportion of social magnifiers of about 22% in the first session and of 15% in the second session. In addition, we find that about 88% of subjects exhibit self bias: they put a larger weight on themselves than on any of their matches.

obtained in our original experiment (reported in Figure 7), we see that the distribution of $\delta$ is more right-skewed in the extension experiment than in the original experiment. Comparing the moments of the distribution of $\delta$ in Table 5 with those in Table 2 confirms this observation. This means that social discounting is steeper in the extension experiment compared to the original experiment.

Moreover, we observe again that, while most of the distribution is below 1 corresponding to social discounting, there is a significant proportion of the distribution above 1 corresponding to social magnification. We find that about 25% of subjects are social magnifiers in the case of the quasi-hyperbolic social discount function, and about 28% in the case of the quasi-hyperbolic quadratic social discount function.[26]

From Figure 8, we also see that the distribution of $\beta$ is mostly below 1, which confirms self bias. We formalise this argument in Table 5 which also reports the moments of the distribution of $\beta$ and hypothesis tests against 1, as we did for $\delta$ in Table 5. We see that the median value of $\beta$ is 0.67 in the quasi-hyperbolic case and 0.70 in the quasi-hyperbolic quadratic case. We reject the null hypothesis that $\beta = 1$ for both models and for all moments of $\beta$.

We compute the dissimilarity premium as in the original experiment, from each subject's posterior means with $x = 1$ and $s = 0.7$, and separately for social discounters and social magnifiers. For social discounters, we obtain a dissimilarity premium of about \$0.62 (95% confidence interval: \$0.57–\$0.66) in the quasi-hyperbolic case and of about \$0.55 (95% confidence interval: \$0.51–\$0.60) in the quasi-hyperbolic quadratic case. For social

[26] As in Section 5.4, for the quasi-hyperbolic quadratic social discount function the proportion of magnifiers is obtained via the joint $\delta$, $\kappa$ procedure described in footnote 21. The $\beta$ self-bias term does not affect this procedure since it multiplies $D(s_{i1})$ and $D(s_{i3})$ equally and so cancels out of the comparison $D(s_{i3}) > D(s_{i1})$.

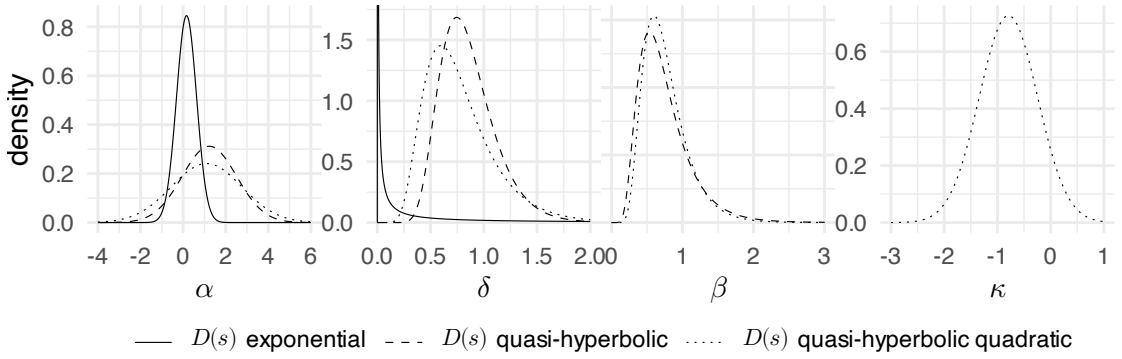



**Figure 8:** Plots of the distributions estimated in Table 4 of $\alpha$ (utility of money curvature), $\delta$ (social discount factor), $\beta$ (self bias parameter), and $\kappa$ (quadratic parameter), extension experiment.

**Table 4:** Maximum simulated likelihood estimates, extension experiment.

| $D(s)$ | No discounting | Exponential | Quasi-hyperbolic | Quasi-hyperbolic quadratic |
|---|---|---|---|---|
| $\mu_\alpha$ | $0.90^{***}$ | $0.16^{***}$ | $1.23^{***}$ | $1.10^{***}$ |
| | (0.11) | (0.01) | (0.13) | (0.07) |
| $\sigma_\alpha$ | $1.26^{***}$ | $0.47^{***}$ | $1.28^{***}$ | $1.65^{***}$ |
| | (0.15) | (0.04) | (0.14) | (0.11) |
| $\mu_\delta$ | | $-12.22^{***}$ | $-0.20^{***}$ | $-0.34^{***}$ |
| | | (0.76) | (0.04) | (0.05) |
| $\sigma_\delta$ | | $17.06^{***}$ | $0.30^{***}$ | $0.42^{***}$ |
| | | (1.05) | (0.05) | (0.06) |
| $\mu_\beta$ | | | $-0.40^{***}$ | $-0.35^{***}$ |
| | | | (0.06) | (0.03) |
| $\sigma_\beta$ | | | $0.47^{***}$ | $0.40^{***}$ |
| | | | (0.06) | (0.03) |
| $\mu_\kappa$ | | | | $-0.79^{***}$ |
| | | | | (0.08) |
| $\sigma_\kappa$ | | | | $0.55^{***}$ |
| | | | | (0.10) |
| $\sigma$ | $0.33^{***}$ | $0.17^{***}$ | $0.07^{***}$ | $0.07^{***}$ |
| | (0.03) | (0.02) | (0.01) | (0.01) |
| Log Likelihood | $-4933.07$ | $-4108.85$ | $-3209.97$ | $-3103.93$ |

*Notes.* Robust standard errors in parentheses.
$\alpha$ and $\kappa$ normally distributed, $\beta$ and $\delta$ log-normally distributed.
$u(x)$ exponential as defined in (1).
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

**Table 5:** Moments of the distributions of $\delta$ and $\beta$ and hypothesis tests for the models reported in Table 4, extension experiment.

| $D(s)$ | Moment | $\delta$ | | | $\beta$ | | |
|---|---|---|---|---|---|---|---|
| | | Estimate | Test against 1 | | Estimate | Test against 1 | |
| | | | $\chi^2$ | $p$ | | $\chi^2$ | $p$ |
| Exponential | mean | $8.01\times10^{57}$ | 0.00 | 0.95 | | | |
| Exponential | median | 0.00 | $7.26\times10^{10}$ | $< 0.001$ | | | |
| Exponential | mode | 0.00 | $2.17\times10^{260}$ | $< 0.001$ | | | |
| Quasi-hyperbolic | mean | 0.86 | 69.05 | $< 0.001$ | 0.75 | 67.60 | $< 0.001$ |
| Quasi-hyperbolic | median | 0.82 | 36.74 | $< 0.001$ | 0.67 | 60.41 | $< 0.001$ |
| Quasi-hyperbolic | mode | 0.75 | 23.72 | $< 0.001$ | 0.53 | 56.70 | $< 0.001$ |
| Quasi-hyperbolic quadratic | mean | 0.78 | 103.64 | $< 0.001$ | 0.76 | 231.19 | $< 0.001$ |
| Quasi-hyperbolic quadratic | median | 0.71 | 55.09 | $< 0.001$ | 0.70 | 196.36 | $< 0.001$ |
| Quasi-hyperbolic quadratic | mode | 0.60 | 39.13 | $< 0.001$ | 0.60 | 163.79 | $< 0.001$ |

*Notes.* Tests are two-sided Wald tests based on robust standard errors.

magnifiers, we obtain a dissimilarity premium of about \$0.38 (95% confidence interval: \$0.22–\$0.52) in the quasi-hyperbolic case and of about \$0.30 (95% confidence interval: \$0.10–\$0.49) in the quasi-hyperbolic quadratic case.[27] Those numbers are larger than the dissimilarity premia we obtained in our original experiment, which reflects the additional effect of self bias.

In Table 6 we compare the goodness of fit of these models. Given the results we have seen so far it is thus not surprising that Table 6 confirms that the quasi-hyperbolic model fits the data better than the exponential model. Table 6 also indicates that the addition of a quadratic term to the quasi-hyperbolic social discount function leads to a better fit, which is again evidence of non-monotonicity of the social discount function. However, this time, the increase in fit resulting from adding this parameter is modest and smaller than the increase in fit resulting from adding the parameter to capture self bias.

[27] The confidence intervals are obtained by bootstrapping subjects within each type, as in the original experiment. Since subjects face different matches in the two sessions, we classify them using the average of their dissimilarity indices $s_{i1}$ and $s_{i3}$ across the two sessions. The means exclude 3 (quasi-hyperbolic) and 6 (quasi-hyperbolic quadratic) of the 109 subjects whose dissimilarity premium is not defined for some $s$ between 0 and 0.9. Pooling both types, the mean dissimilarity premium is about \$0.57 in the quasi-hyperbolic case and \$0.53 in the quasi-hyperbolic quadratic case.

**Table 6:** Comparison of the fit of the models estimated in Table 4.

| $D(s)$ | Log-likelihood | Likelihood-ratio test | | Hit rate |
|---|---|---|---|---|
| | | $\chi^2$ | $p$ | |
| No discounting | −4933.07 | | | 0.57 |
| Exponential | −4108.85 | 1648.43 | $< 0.001$ | 0.71 |
| Quasi-hyperbolic | −3209.97 | 1797.77 | $< 0.001$ | 0.79 |
| Quasi-hyperbolic quadratic | −3103.93 | 212.07 | $< 0.001$ | 0.80 |

*Notes.* The likelihood-ratio test compares the model in one row to the model in the row above it.
Hit rate is the proportion of choices predicted correctly by the model.

# 7 Robustness checks and additional analyses

## 7.1 Preference for more dissimilar matches

From the previous sections, we have obtained three key results: a majority of subjects are social discounters; about 30% of subjects are social magnifiers who prefer to give money to those less similar; and the social discount function is not necessarily monotone.

It might seem surprising that some subjects are social magnifiers and prefer more dissimilar matches. In this subsection, we present additional analyses to check the robustness of this result. We focus here on the data from the original experiment for simplicity. We find qualitatively similar results when we use data from the extension experiment or when we pool the data from both experiments.

### 7.1.1 Ordering of the matches

A potential explanation to the prevalence of social magnifiers is that the dissimilarity index does not accurately capture how subjects perceive the social distance between themselves and their matches. For example, looking at Figure 5, we see that the second most common ordering is $d_{i1} > d_{i3} > d_{i2}$. Recall that the dissimilar match is indexed by 3 and thus corresponds to $d_{i3}$. However, if subjects actually perceived the intermediate match to be more socially distant than the most dissimilar match, then the intermediate match would be indexed by 3 and the dissimilar match would be indexed by 2. As a consequence, the order of intermediate and dissimilar matches would be flipped and the true ordering would become $d_{i1} > d_{i2} > d_{i3}$, which would therefore be consistent with monotone social discounting.

To test this hypothesis, we substitute the dissimilarity index with the overlaps from the Continuous IOS scale to order the matches. The Continuous IOS scale directly measures

how subjects perceive the social distance between themselves and their matches. When we order them with the Continuous IOS overlap they reported for each match, 1 now corresponds to the match that has received the highest overlap and 3 to the match that has received the lowest overlap. According to the IOS scale, high overlap indicates high social connection and thus low social distance. We then repeat the analysis of Section 5.3. If social magnification was really a false artefact of the use of the dissimilarity index, then social magnifiers should disappear.

We report the orderings in Table 7. As we can see by comparing this table with Figure 5, the orderings appear in relatively the same order and occur in roughly the same proportions. In particular, the orderings that place the dissimilar match above some other matches, for example $d_{i1} > d_{i3} > d_{i2}$ or $d_{i3} > d_{i2} > d_{i1}$, appear in essentially the same spots on the list. Overall, using the IOS-based ordering, about 22% of subjects are still classified as social magnifiers.

This finding suggests that observing subjects who prefer more dissimilar matches is not an artefact of the dissimilarity index not correctly ordering matches. More socially distant matches are genuinely preferred by some subjects as revealed by the Continuous IOS scale.

### 7.1.2 Redistributive concerns

Another possible explanation for the finding that some subjects are social magnifiers is that these preferences reveal redistributive concerns. According to this explanation, perhaps some subjects prefer the dissimilar match because they perceive the dissimilar match to be worse off and wish to redistribute.

We were well aware of this issue when designing the experiment. For this reason, we were careful not to include any variables on the card displays that would trigger redistributive concerns. For example, we did not include any information about the matches' income, their employment status, or how many children they have. However, our decision to not include these variables explicitly does not mean that subjects did not use the information we did include to implicitly form beliefs about them. For example, a subject facing a dissimilar match who is Black might believe that this match is financially worse off than another on the basis of race alone, and thus prefer to give money to them.

By potentially increasing the weight placed on the most dissimilar match, redistributive concerns run counter to social discounting. Therefore, monotone social discounters should exhibit even more social discounting if there are redistributive concerns and we control for these. That said, redistributive concerns cannot explain every unexpected ordering, but

**Table 7:** Orderings resulting from $d_{ij}$ estimated, matches ordered by Continuous IOS overlap.

| Ordering | Type | Shape | $n$ | % | Cumulative % |
|---|---|---|---|---|---|
| $d_{i1} > d_{i2} > d_{i3}$ | social discounter | monotone | 124 | 34.93 | 34.93 |
| $d_{i1} > d_{i3} > d_{i2}$ | social discounter | U-shaped | 50 | 14.08 | 49.01 |
| $d_{i2} > d_{i1} > d_{i3}$ | social discounter | inverse U-shaped | 49 | 13.80 | 62.82 |
| $d_{i1} = d_{i2} = d_{i3}$ | no discounting | | 33 | 9.30 | 72.11 |
| $d_{i3} > d_{i1} > d_{i2}$ | social magnifier | U-shaped | 29 | 8.17 | 80.28 |
| $d_{i2} > d_{i3} > d_{i1}$ | social magnifier | inverse U-shaped | 23 | 6.48 | 86.76 |
| $d_{i3} > d_{i2} > d_{i1}$ | social magnifier | monotone | 23 | 6.48 | 93.24 |
| $d_{i1} > d_{i2} = d_{i3}$ | social discounter | | 11 | 3.10 | 96.34 |
| $d_{i1} = d_{i2} = d_{i3} = 0$ | indifferent in all choices | | 6 | 1.69 | 98.03 |
| $d_{i3} > d_{i1} = d_{i2}$ | social magnifier | | 4 | 1.13 | 99.15 |
| $d_{i2} > d_{i1} = d_{i3}$ | | | 2 | 0.56 | 99.72 |
| $d_{i1} = d_{i2} > d_{i3}$ | social discounter | | 1 | 0.28 | 100.00 |
| Total | | | 355 | 100 | |

only those in which some subjects prefer more dissimilar matches such as, for example, U-shaped social discounters or social magnifiers.

To address the possibility of redistributive concerns, we augment our data with estimates of the income of the matches. The estimates come from the American Community Survey (ACS) 2019 5-year data. The ACS generates separate estimates of median earnings by gender for race and ethnicity in every zip code in the United States. These variables—gender, race, ethnicity, and location—were listed on the card display and, therefore, subjects could use them to form their own inferences on their matches' income. To integrate this ACS income data into our analysis, it was necessary to make some assumptions, the details of which are given in Appendix E where we also more fully explain how we integrate the ACS data into our analysis.

Once we have augmented our data with the income of the matches, we estimate a mixed logit model with a linear form. The use of this model will allow us later to easily include the income of the match as an additional independent variable. We assume that all parameters are normally distributed.

Table 8 reports the estimation results.[28] In Model (1), we add money amount and

[28]The Table only reports 349 subjects instead of the full 355 because the remaining subjects reported indifference in all their choices and so cannot be used by the mixed logit model. In addition, we do not have the full 27 choices for all subjects because we could not always recover an estimated income for each match: the ACS estimates of median earnings are not available for every combination of sex,

dissimilarity index as independent variables, essentially replicating our previous analysis with a reduced-form linear model.

All the results we found previously hold. The estimate of the mean of dissimilarity is negative and significant, which implies that on average subjects are social discounters. The estimate of the standard deviation of dissimilarity is also significant, which means that there is heterogeneity in the degree of social discounting across subjects. From the mean and standard deviation estimates, we can compute that about 33% of subjects are social magnifiers—again, in line with our previous results.

The real test of redistributive preferences starts with Model (2) in which we add the estimated income of the match as an additional independent variable. The estimated mean of the coefficient on the income of the match is negative but not significant, so on average we find no evidence that subjects prefer to give money to matches with a lower income. The estimated standard deviation is large and significant, however, which means that subjects differ in how they respond to the income of their match. What matters most for our purposes is that the estimated mean of dissimilarity remains negative and significant. If anything, it becomes more negative once the income of the match is included, meaning that subjects exhibit even more social discounting as we control for the income of the match. In this model, we estimate that about 31% of subjects are social magnifiers.

Models (3) and (4) perform the same analyses as Models (1) and (2), but they add a quadratic term for the dissimilarity index as an additional independent variable. Here the coefficient on the income of the match is negative and significant, so these specifications do provide some evidence of redistributive concerns. Our main result is unaffected: the mean of dissimilarity remains large, negative, and significant in both models. The share of social magnifiers is unaffected as well: we obtain about 42% for Model (3) and about 34% for Model (4).[29] Therefore, while there is some evidence of redistributive concerns, our main results do not change when we control for them.

race, ethnicity, and zip code (see Appendix E). When a match's estimated income is missing, all choices involving that match are dropped from the estimation sample. In total we drop only 495 choices out of 8452 (the remaining non-indifferent choices), and dropping these choices only affects 30 subjects. We apply this restriction to all four models in Table 8 so that all four columns are estimated on the same sample and are directly comparable.

[29] Applying the procedure described in footnote 21, the condition to be a magnifier is $\beta_1 + \beta_2(s_{i1} + s_{i3}) > 0$, where $\beta_1$ and $\beta_2$ are the coefficients on dissimilarity and dissimilarity squared. We evaluate the resulting probability at each subject's own $(s_{i1}, s_{i3})$ and average across subjects.

**Table 8:** Effect of income. Mixed logit model.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Means: | | | | |
| Money amount | $1.644^{***}$ | $2.459^{***}$ | $1.946^{***}$ | $2.357^{***}$ |
| | (0.129) | (0.193) | (0.158) | (0.167) |
| Dissimilarity | $-1.598^{***}$ | $-2.899^{***}$ | $-8.332^{***}$ | $-8.177^{***}$ |
| | (0.233) | (0.368) | (0.571) | (0.638) |
| Dissimilarity squared | | | $7.247^{***}$ | $7.241^{***}$ |
| | | | (0.709) | (0.907) |
| Estimated income | | −0.022 | | $-0.222^{**}$ |
| | | (0.060) | | (0.089) |
| Standard deviations: | | | | |
| Money amount | $1.706^{***}$ | $2.416^{***}$ | $2.640^{***}$ | $2.326^{***}$ |
| | (0.140) | (0.180) | (0.360) | (0.160) |
| Dissimilarity | $3.679^{***}$ | $5.735^{***}$ | $9.792^{***}$ | $4.044^{***}$ |
| | (0.221) | (0.357) | (0.816) | (0.436) |
| Dissimilarity squared | | | $10.924^{***}$ | $5.439^{***}$ |
| | | | (0.931) | (0.382) |
| Estimated income | | $1.450^{***}$ | | $1.342^{***}$ |
| | | (0.094) | | (0.101) |
| Log Likelihood | −3396.041 | −3075.861 | −3133.161 | −2985.850 |
| Number of subjects | 349 | 349 | 349 | 349 |
| Number of choices | 7957 | 7957 | 7957 | 7957 |
| Number of observations | 15914 | 15914 | 15914 | 15914 |

*Notes.* $^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.
Coefficients for which we report the standard deviation are assumed to be normally distributed.
Income expressed in tens of thousands of dollars.

## 7.2 Alternative utility function and distributional assumption

So far our parametric results rest on the assumption of an exponential utility function and a log-normal distribution of $\delta$. Here we consider alternative specifications and show again that our results remain robust to these alternatives.

In terms of data, we pool all the data we have available, meaning the data from the original experiment (Section 4) and the data from the extension (Section 6). For the extension, we also include data from subjects who only completed the first session and for whom we have 27 observations instead of the full 81.

For $u(x)$, in addition to the exponential utility function, we also consider the power utility function. For the distribution of $\delta$, in addition to the lognormal distribution, we also consider the normal distribution left-truncated at 0. For the social discount function $D(s)$, we focus on the quasi-hyperbolic and quasi-hyperbolic quadratic forms which allow for self bias. Because the pooled data include the extension data, in which subjects trade off money for themselves against money for others, we do not consider social discount functions without self bias. We estimate all combinations of these specifications meaning that we estimate 2 (social discount functions) $\times$ 2 (utility functions) $\times$ 2 (distributions of $\delta$), for a total of 8 models.

We report detailed results in Appendix F. We focus here on the distribution of $\delta$ across models which we report in Figure 9. As we can see in this figure, the distribution of $\delta$ is broadly stable across these specifications. Consequently, our results carry over to these alternative specifications.

First, the distribution of $\delta$ is generally between 0 and 1 across models, which is consistent with social discounting. Second, it remains the case that there is a significant proportion of the distribution of $\delta$ above 1 across models which is consistent with social magnification.

In the Appendix we report the moments of the distribution of $\delta$ and hypothesis tests for all models as well as comparisons of the fit of all models. Our results are in line with what we have reported so far. The models with the assumptions we have made so far—exponential utility function, lognormal distribution of $\delta$—are among the best fitting models.

## 7.3 Social discounting without revealing demographic information

Finally, we show that we can replicate our analysis using data from a setting in which subjects were not provided the demographic information of their matches. This analysis

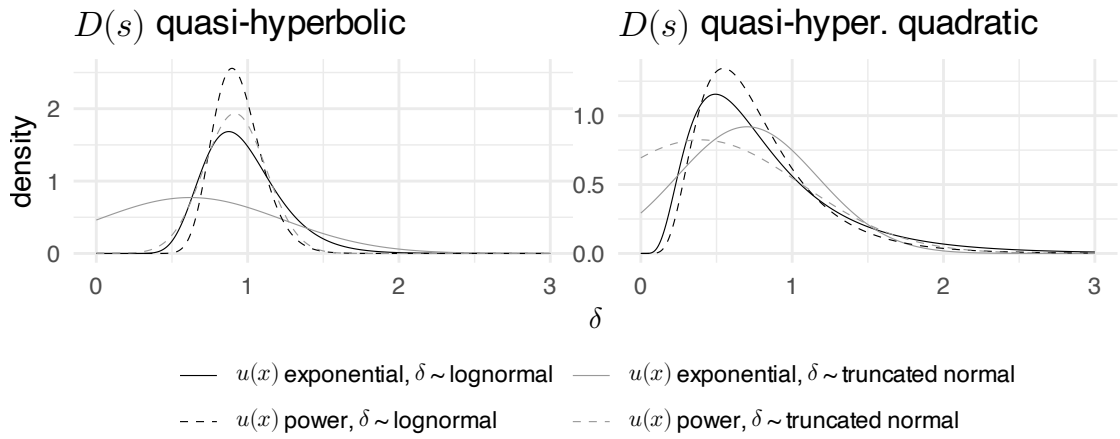



**Figure 9:** Plots of the distributions for the models reported in Appendix F for $\delta$ (social discount factor), pooled data.

addresses the concern that providing subjects the demographic information of their matches rendered similarity too salient.

The setting comes from Robson (2021), who conducted a lab-in-the-field experiment in Uganda. We refer the reader to Robson (2021) for a detailed description of the experiment giving only a brief overview of the design here.

### 7.3.1 Design of Robson (2021)

In this experiment, subjects play 27 rounds of a three-person dictator game. In each round, subjects divide a budget of 60,000 Ugandan schillings as Player 1 between themselves and two other subjects, Players 2 and 3. Between each round, a multiplier changes for each player, in a manner similar to Andreoni and Miller (2002) and Fisman et al. (2007): an amount allocated to a player is converted into a payoff for that player by multiplying it by the player's multiplier, so a higher multiplier made giving to that player more efficient. Players 2 and 3 change between each round. In the treatment that interests us—the 'Known' treatment—subjects could see each other, so dictators could identify Players 2 and 3 when making their allocation decisions.

As can be seen from this short description, not providing the demographic information to subjects is not the only difference between this experiment and our own. Robson (2021) took place in the field as opposed to an online lab setting. The subject pool is also different. Further, it uses a different task—a three-person dictator game with a continuous choice set—as opposed to our binary choice task. Showing that our results replicate in this different setting thus allows us to establish greater external validity for our results.

The demographic characteristics measured by Robson (2021) were: gender, age, education, religion, and tribe. We use these five variables to compute a dissimilarity index like the one we computed in our experiments. To be able to compare estimates with those we obtained from our experiments, we convert the payoffs into US dollars using the exchange rate at the time of the experiment (Robson, 2021, footnote 8). As a result of this conversion, the total amount to divide becomes \$18. If subjects split the money equally, each recipient would receive \$6, which puts the payoffs in the same range as in our experiments.

### 7.3.2 Functional form and stochastic assumptions

We estimate the same type of model as we did for our experiments. For the social discount function $D(s)$ we use the quasi-hyperbolic quadratic form since it was the one resulting in the best fitting model. For the utility function of money $u(x)$ we choose the power utility function

$$u(x, \alpha_i) = \begin{cases} x^{\alpha_i} \text{ if } \alpha_i > 0, \\ \log(x) \text{ if } \alpha_i = 0, \\ -x^{\alpha_i} \text{ if } \alpha_i < 0, \end{cases} \tag{3}$$

where $\alpha_i$ captures the curvature: $u(x)$ is concave for $\alpha_i < 1$, convex for $\alpha_i > 1$, and linear for $\alpha_i = 1$. This utility function is closer to the one used by Robson (2021) than the exponential utility function. Using the exponential utility function leads to the same qualitative results as those we report below at the cost of a worse fit.

Denote by $x_{ij}$ the allocation of subject $i$ to recipient $j$, and by $p_j$ recipient $j$'s multiplier, so that recipient $j$'s payoff is $p_j x_{ij}$. Subject $i$ chooses $\{x_{ij}\}_{j=1}^3$ to maximise $\sum_{j=1}^3 D(s_{ij}; \beta_i, \delta_i, \kappa_i)\, u(p_j x_{ij}, \alpha_i)$ subject to the budget constraint $m = \sum_{j=1}^3 x_{ij}$ and $x_{ij} \geq 0$. Denote by $x_{ij}^*$ the resulting optimal allocation. For $\alpha_i < 1$, we have

$$x_{ij}^*(\alpha_i, \beta_i, \delta_i, \kappa_i) = \frac{m}{\sum\limits_{k=1}^{3} \left( \frac{D(s_{ik}; \beta_i, \delta_i, \kappa_i)}{D(s_{ij}; \beta_i, \delta_i, \kappa_i)} \right)^{\frac{1}{1-\alpha_i}} \left( \frac{p_k}{p_j} \right)^{\frac{\alpha_i}{1-\alpha_i}}}.$$

This is the same closed-form solution as in Robson (2021)'s own CES model, up to a change of parametrisation.

To take the model to the data, we make the same stochastic assumption as Robson (2021): the observed allocations $\{x_{ij}\}_{j=1}^3$ are drawn from a Dirichlet distribution with

mean equal to the optimal allocation $\{x^*_{ij}\}_{j=1}^3$ and a precision parameter $\varsigma$, common to all subjects, such that the Dirichlet shape parameters are $a_{ij} = (x^*_{ij}/m)(\varsigma - 1)$, with $\varsigma > 1$. We refer the reader to Robson (2021) for additional details.[30]

Since the model requires $\alpha_i < 1$, as in Robson (2021), we assume that $1 - \alpha_i$ is log-normally distributed, $\ln(1 - \alpha_i) \sim \mathcal{N}(\mu_\alpha, \sigma^2_\alpha)$. All other aspects of the estimation remain the same: we assume that the parameters are drawn from distributions and we use the method of maximum simulated likelihood to estimate the moments of these distributions.

### 7.3.3 Estimation results

We report the estimation results in Table 9. In Figure 10, we plot these estimated distributions and compare them to the estimated distributions obtained when using the pooled data from all our experiments and the same parametric assumptions ($u(x)$ power, $D(s)$ quasi-hyperbolic quadratic; see Section 7.2 and Appendix F).

We see that our results generally carry over to the data from Robson (2021): we observe heterogeneity in the social discount factor $\delta$, a non-negligible proportion of social magnifiers, and self bias. However, we also observe some differences. We first see that the estimated distribution of $\beta$ is closer to 0 in Robson (2021) than in our experiments, which means that in Robson (2021) subjects exhibit more self bias. Further, in Robson (2021), the distribution of $\delta$ has more mass above 1, which means that there are even more social magnifiers in Robson (2021) than in our experiments. Finally, we see that the additional quadratic term $\kappa$ is generally lower in Robson (2021), which means that there are more subjects with non-monotonous social discount functions compared to our experiments. In Appendix G we show that the nonparametric estimation of the social discount function leads to the same conclusions.

# 8 Discussion and conclusion

We have reported the results from two experiments in which we used a continuous measure of social distance—interpersonal similarity—to manipulate social distance. We have used social discounting as a structural model and have carried out both nonparametric and

[30] In particular, the shape parameters $a_{ij}$ need to be strictly positive, which requires the optimal allocations $x^*_{ij}$ to be strictly positive. This holds automatically once $\alpha_i < 1$, since the closed-form solution above never predicts a zero allocation. The observed allocations $x_{ij}$, however, can be exactly zero in the data. For this reason, we follow Fisman et al. (2007) and Robson (2021) and add an epsilon ($10^{-6}$) to any zero-valued allocation.

**Table 9:** Maximum simulated likelihood estimates, data from Robson (2021).

| | Estimate | Standard error |
|---|---|---|
| $\mu_\alpha$ | 0.09 | (0.10) |
| $\sigma_\alpha$ | 1.14 | (0.09)$^{***}$ |
| $\mu_\beta$ | −1.05 | (0.18)$^{***}$ |
| $\sigma_\beta$ | 2.57 | (0.35)$^{***}$ |
| $\mu_\delta$ | 0.47 | (0.15)$^{**}$ |
| $\sigma_\delta$ | 0.92 | (0.11)$^{***}$ |
| $\mu_\kappa$ | −1.55 | (0.44)$^{***}$ |
| $\sigma_\kappa$ | 0.61 | (0.26)$^{*}$ |
| $\varsigma$ | 10.11 | (2.48)$^{***}$ |
| Log Likelihood | 7854.44 | |

*Notes.* Robust standard errors.
$D(s)$ quasi-hyperbolic quadratic, $u(x)$ power as defined in (3). $\alpha$ is distributed as one minus a log-normal random variable, so that $\alpha < 1$; $\beta$ and $\delta$ are log-normally distributed; $\kappa$ is normally distributed.
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

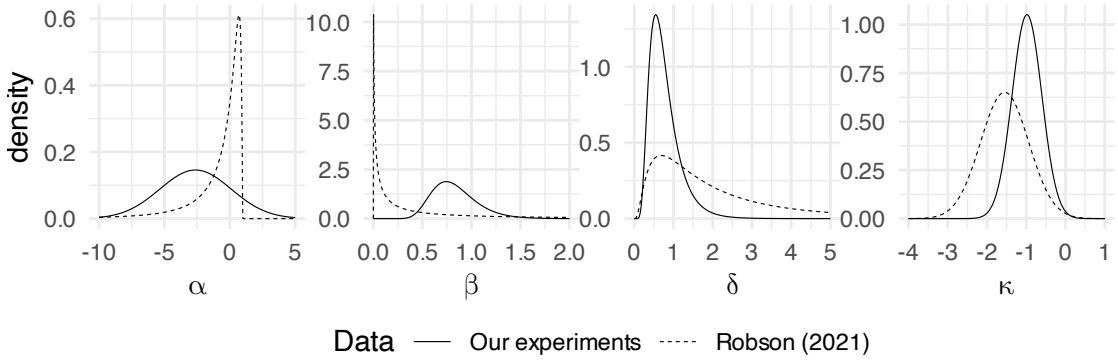



**Figure 10:** Comparisons of the estimated distributions of $\alpha$ (utility of money curvature, bounded above by 1 for Robson (2021)), $\beta$ (self bias parameter), $\delta$ (social discount factor), and $\kappa$ (quadratic parameter), between Robson (2021) (Table 9) and our experiments.

parametric analyses. In line with the literature, we have found that most subjects are social discounters, meaning that they prefer socially closer others to socially farther others. However, we have also found that a significant proportion of subjects are social magnifiers, meaning that they prefer socially farther others. Our results hold after we perform a number of robustness checks.

While the finding that some people are social magnifiers may be surprising at first glance, we can find echoes of it elsewhere. For example, Buchan et al. (2006) find that, contrary to the usual finding of ingroup bias, Chinese subjects exhibit outgroup favouritism meaning that they prefer money for outgroup members to money for ingroup members. Further, there are parallels in the intertemporal choice literature: we might think that everyone discounts the future, but there is evidence that some people prefer the future. Takeuchi (2011), for instance, find that future-biased preference reversals significantly outnumber present-biased ones.[31] Beyond science and in the realm of literature, this also seemed to be well-known to Dostoevsky who, in *The Brothers Karamazov*, describes Ivan Fyodorovich as a social magnifier:

> I could never understand how one can love one's neighbours. It's just one's neighbours, to my mind, that one can't love, though one might love those at a distance. For any one to love a man, he must be hidden, for as soon as he shows his face, love is gone. [...] And what's more, a man is rarely ready to admit another's suffering (as though it were a distinction). Why won't he admit it, do you think? Because I smell unpleasant, because I have a stupid face, because I once trod on his foot. [...] Beggars, especially genteel beggars, ought never to show themselves, but to ask for charity through the newspapers. One can love one's neighbours in the abstract, or even at a distance, but at close quarters it's almost impossible.

This quote also provides some clues as to why some people are social magnifiers: socially close others, and their human flaws, are more easily identifiable and thus more likely to be the object of negative feelings such as resentment, jealousy, or envy.

Throughout the paper, we have highlighted connections with the intertemporal choice literature. That is by choice and not accident, as research in psychology shows that the way people deal with social distance is related to the way they deal with time.[32] Therefore, just like there is a literature connecting risk preferences and time preferences

[31] See also Loewenstein (1987), Sayman and Öncüler (2009), Meier and Sprenger (2010), Jackson and Yariv (2014), Montiel Olea and Strzalecki (2014), and Balakrishnan et al. (2020) for related evidence.

[32] This research is mostly part of the construal-level theory literature. See, for example, Pronin and Ross (2006), Pronin et al. (2008), Bartels and Rips (2010), Stephan et al. (2011), and Yi et al. (2011).

(Epper and Fehr-Duda, 2024; Epper et al., 2011; Halevy, 2008; Prelec and Loewenstein, 1991; Quiggin and Horowitz, 1995), we expect that time preferences and social distance preferences might also be related. In an experiment, one could measure time preferences using an established method (Andreoni et al., 2015) and use our method to measure social distance preferences. We would predict that the time discount rate is related to the social discount rate.

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# Similarity and social discounting: Measuring preferences for social distance

Online Appendix

## Appendix A Experimental materials

This Appendix reports the experimental materials.

When they joined the experiment, subjects started by completing the forty question survey described in Section 4.1 'Stage 1'. The list of questions used in the survey appears in the Appendix of Beranek and Castillo (2024), which relied on the same survey.

Then, they went through the following tasks. Subjects always completed the matches display task (Section A.1.1) first. Then, half of the subjects faced the continuous IOS scale task (Section A.1.2) before the choice task (Sections A.1.3 and A.1.4), and the other half faced the choice task before the continuous IOS scale task. The section titles were not shown to subjects.

Section A.1 reports the instructions that subjects saw in the experiment. Section A.2 shows screenshots of the tasks as subjects saw them.

### A.1 Instructions

#### A.1.1 Matches display

Previously, we invited several hundred participants to complete the same survey as you. Out of that group, we selected 3 participants and display some of their responses here.

In order to more easily refer to them we have randomly associated each with one of the four suits from a deck of playing cards: Spade (♣), Club (♠), Heart (♥), and Diamond (♦).

We will represent these participants and some of their answers to the survey with the following cards:

*[Figure A1]*

For the rest of the study, you will complete tasks involving these 3 participants and they will remain the same throughout.

In the next pages we will ask you to consider these participants one by one and write a few words about them.

### A.1.2 Continuous Inclusion of Other in the Self scale

We will now ask you to consider how connected you feel towards the participants.

More specifically, we will show you the following circles:

*[Figure A2]*

You can manipulate the circles by clicking on the left circle, then drag-and-dropping it.

The overlap of the circles represents the connection you feel towards the other participant. For example, a stronger connection between you and the other participant would be represented by a greater overlap of the circles.

We will ask you to shift the circles until you find a configuration that best represents the connection you feel toward each of the 3 participants whose survey responses you just read.

You can practice on this page as much as you want.

To go to the next page, please change the circles so that they're overlapping as much as possible. Once you've done so the 'Next' button will appear.

### A.1.3 Choice task

According to your decisions in this task, we will send money to one of the participants whose survey answers you just read. You are literally picking who receives money from us.

Recall that these participants were selected from several hundred MTurk Workers who previously completed the same survey as you.

The amount of money the MTurker receives will be \$1.00, \$2.00, or \$5.00, and they will receive it in the form of an MTurk bonus payment. This MTurk bonus payment will come from us without explanation. This money is completely independent of the \$3.00 reward you will receive for completing this HIT.

It is therefore important that you read these instructions carefully, because your decisions will determine the bonus payment of one of these MTurk workers.

**Outline of this task** You will be asked 27 questions of the same type. In each question you will be presented with two options and you will be asked to pick the one you prefer.

After you have answered all 27 questions, one of them will be selected at random and the option that you picked will be implemented: if, for example, you picked an option that directed us to give \$1.00 to a particular MTurk worker, we will pay this MTurk worker \$1.00.

Therefore, you should answer each question as if it will be the question that is selected at the end of the experiment to determine another MTurk worker's payment.

**The questions** A sample question is shown below:

*[Figure A3]*

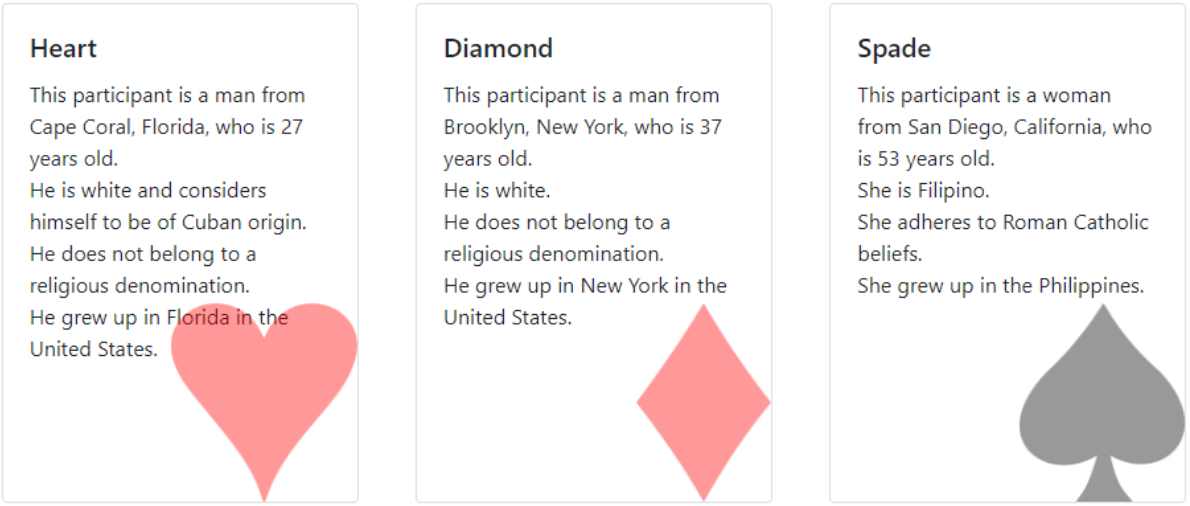



**Figure A1:** Cards presented in the display of the matches.

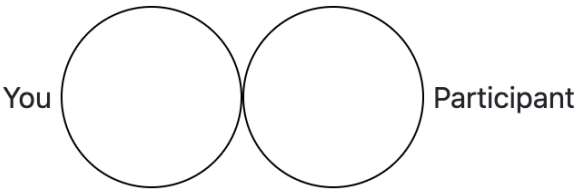



**Figure A2:** Continuous IOS scale.

You can see that there are two options—one on the left and one on the right. The option on the left would lead us to give $2.00 to Heart. The option on the right would lead us to give $5.00 to Diamond.

You have to decide for each question whether you prefer the option on the left or the option on the right. You indicate your choice by clicking on the button below your preferred option. If you feel indifferent between the two options you will also be able to click the button labelled 'Don't care'.

The buttons will appear after 4 seconds and you can spend as much time as you want on each choice.

Since your decisions will affect another MTurk worker you should be as careful as possible when answering the questions.

**The end of the task** After you have answered all 27 questions we will select one at random and will implement what you chose.

For example, if the question above was selected at the end of the experiment and if you had chosen the Left Option, then we would send a bonus payment of $2.00 to Heart. If, on the other hand, you had chosen the Right Option, then we would send a bonus payment of $5.00 to Diamond. Finally, if you had chosen 'Don't care', we will flip a coin to randomly select one MTurk worker to receive all of their allocation.

Therefore, as you go through the questions, remember that each question could be used to determine a real cash payment to another MTurk worker like yourself.

We will maintain your anonymity throughout and the MTurk bonus payment will come from us without explanation.

**Transparency** We are committed to transparency in making these payments and so we will post records of these transactions on our website: `https://geoffreycastillo.com/mturk`.

Three days after the experiment, you will find two documents:

- A Record of the Relevant Decisions from the Experiment that includes
  - the decision situation that was randomly selected
  - the option you chose in that decision situation
  - the partially redacted MTurk worker ID of the participant who we sent the money to
  - your partially redacted MTurk worker ID so you can find this information easily
- A copy of the Complete Transaction History showing payment to the relevant MTurk worker (again with their MTurk worker ID partially redacted)

After we give the money to this MTurk worker, we will send you a bonus of $0.01 with a message reminding you of this procedure and directing you to our website so that you can verify these transactions (remember that this bonus is in addition to your $3.00 reward for completion of the HIT). You can see an example of these records on our website: `https://geoffreycastillo.com/mturk`.

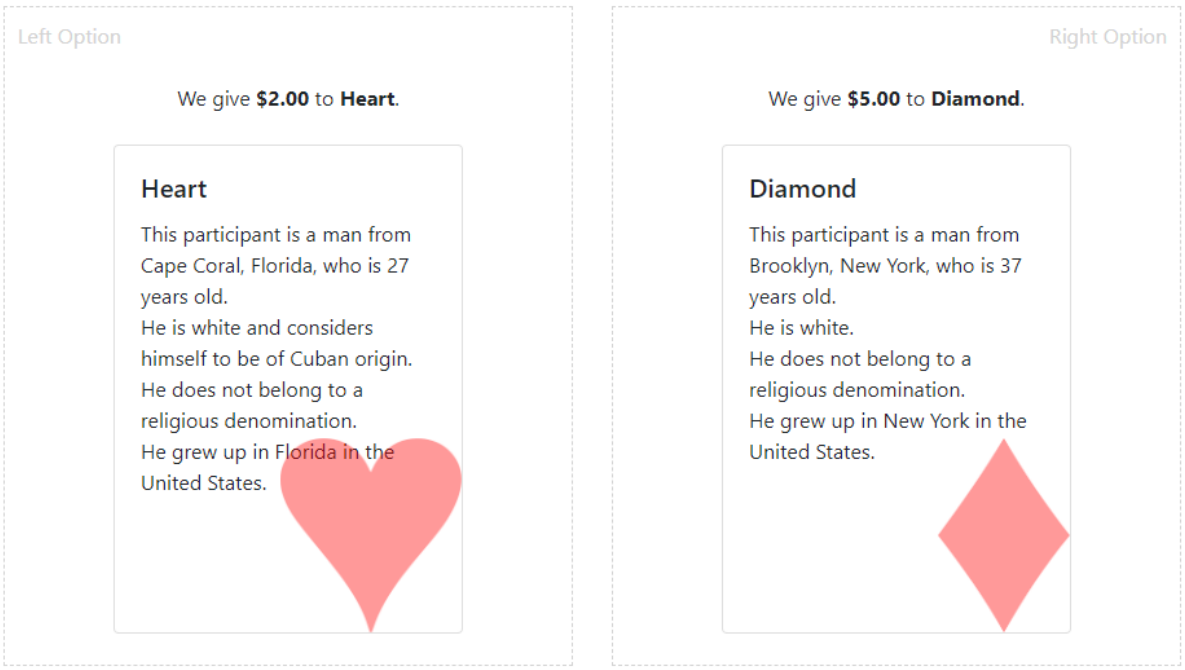



**Figure A3:** Choice task example in the instructions.

### A.1.4 Control questions

To test your understanding we have devised the following control questions.
If you want to take another look at the instructions, click the following button:

[button to reopen instructions]

When I decide an allocation between two participants, those participants are: *[single choice]*

- Hypothetical people who do not exist in real life
- Real MTurk workers who completed the same survey as me, but they won't actually receive any payments
- Real MTurk workers who completed the same survey as me and they will receive payments according to my decisions

When participants receive money as a consequence of my decisions, what will they know? *[single choice]*

- They will know that I made the choice
- They will know that another MTurk worker made the choice, but they won't know it was me in particular
- They won't know anything, as the money will come directly from the researchers and no mention of the present study will be made

True or False: The payment to the participants will come from my own reward. *[single*

*choice]*

- True: The researchers will deduct the payment to the chosen participant from my reward
- False: My reward of $3.00 is independent of the payment to the chosen participant

True or False: Even if I choose 'Don't care' only one participant will receive a payment: *[single choice]*

- True: Choosing 'Don't care' means that one participant is randomly selected to receive all of their allocation
- False: Choosing 'Don't care' means that both participants receive a portion of the allocation

How many of my decisions will be implemented? *[Integer input]*

True or False: I will not be able to verify that any payments have been made: *[single choice]*

- True: I just have to trust the MTurk Requesters will implement my decision
- False: I can verify that payments have been made by visiting `https://geoffreycastillo.com/mturk` three days after the experiment

## A.2 Screenshots of the tasks

This section shows screenshots of the tasks as subjects saw them.

- Figure A4 gives an example of the writing task in which subjects wrote the first things that came to mind about one of their matches.
- Figure A5 gives an example of the Continuous IOS scale task before the subject moved the circles.
- Figure A6 gives an example of a choice in the choice task.

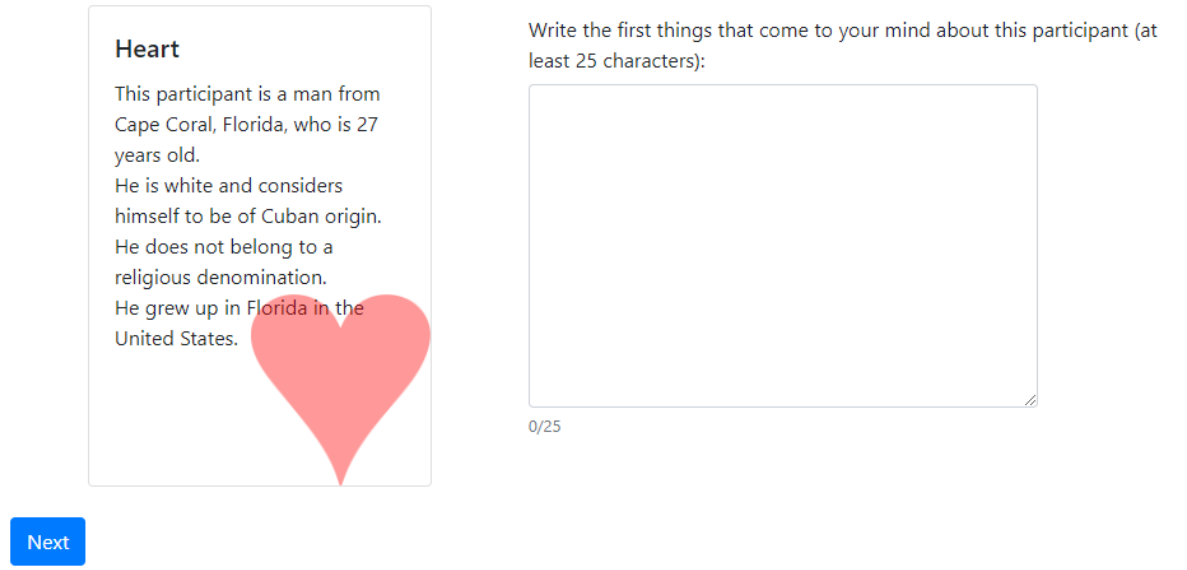



**Figure A4:** Writing task.

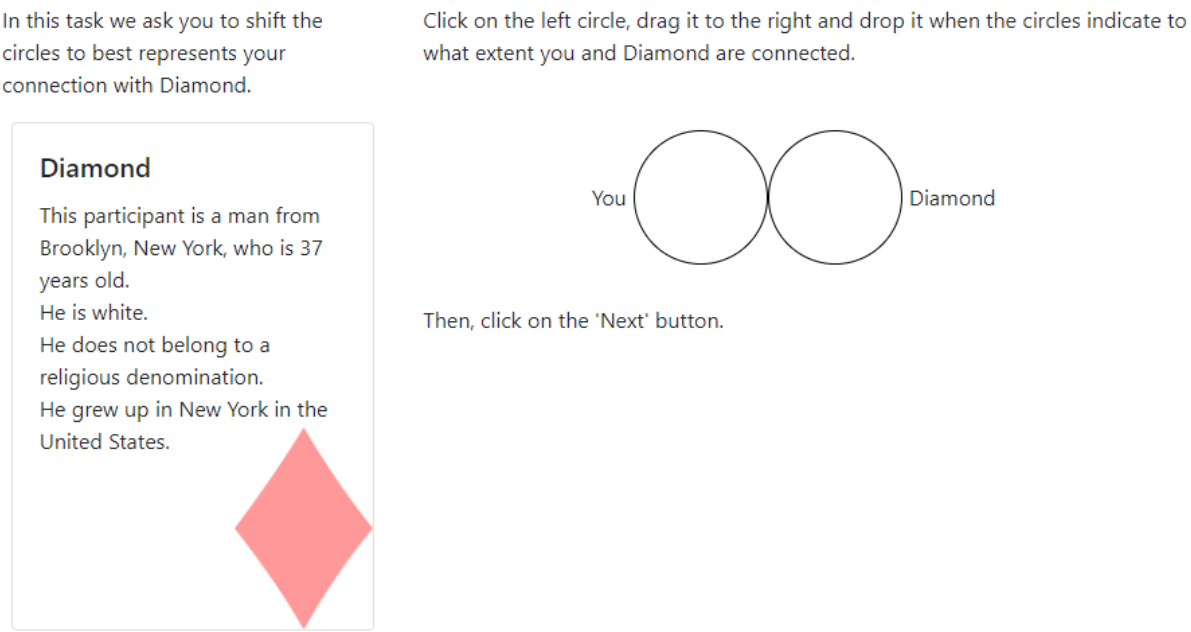



**Figure A5:** Continuous IOS scale task.

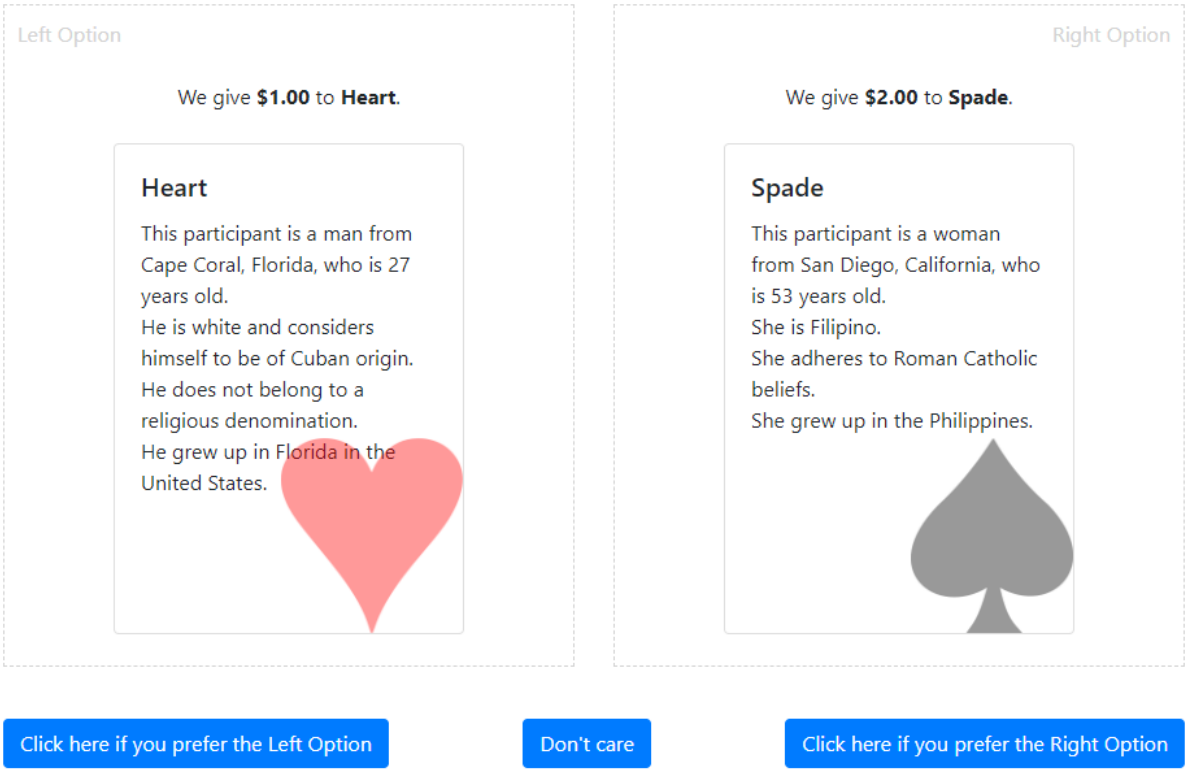



**Figure A6:** Choice task.

# Appendix B Order effects

In our experiment, half of the subjects faced the Continuous IOS scale task before the choice task, and the other half faced the choice task before the Continuous IOS scale task. In this Appendix we show that the order of the tasks did not affect the subjects' choices.

In Table B1 we report the results of panel-data mixed logit models in which we explain choice as a function of the options' attributes, amount of money and dissimilarity. In the first model we restrict to subjects who faced the choice task first, and in the second model we restrict to subjects who faced the Continuous IOS scale task first. In the third model, we combine both orders, identify subjects who did the IOS scale task first, and add interaction effects. As can be seen, we do not find evidence that the order of the tasks affected subjects' choices.

**Table B1:** Panel-data mixed logit model, choice task first or Continuous IOS scale task first.

| | Choice first | IOS first | Combined |
|---|---|---|---|
| Means: | | | |
| Money amount | $1.627^{***}$ | $1.527^{***}$ | $1.618^{***}$ |
| | (0.000) | (0.000) | (0.000) |
| Dissimilarity | $-1.571^{***}$ | $-1.605^{***}$ | $-1.466^{***}$ |
| | (0.000) | (0.000) | (0.000) |
| IOS first=1 × Money amount | | | −0.001 |
| | | | (0.995) |
| IOS first=1 × Dissimilarity | | | −0.114 |
| | | | (0.789) |
| Standard deviations: | | | |
| Money amount | $1.789^{***}$ | $1.576^{***}$ | $1.676^{***}$ |
| | (0.000) | (0.000) | (0.000) |
| Dissimilarity | $3.494^{***}$ | $3.433^{***}$ | $3.412^{***}$ |
| | (0.000) | (0.000) | (0.000) |
| Log Likelihood | −1832.249 | −1802.965 | −3635.782 |
| Number of subjects | 175 | 174 | 349 |
| Number of choices | 4283 | 4169 | 8452 |
| Number of observations | 8566 | 8338 | 16904 |

*Notes.* All coefficients assumed to be normally distributed.
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

# Appendix C Original experiment: detailed table of nonparametric orderings

In the main text, we reported only the six most common orderings based on the $d_{ij}$, which represented about 88% of subjects. We report all the orderings in Table C1.

**Table C1:** Orderings resulting from $d_{ij}$ estimated.

| Ordering | Type | Shape | $n$ | % | Cumulative % |
|---|---|---|---|---|---|
| $d_{i1} > d_{i2} > d_{i3}$ | social discounter | monotone | 97 | 27.32 | 27.32 |
| $d_{i1} > d_{i3} > d_{i2}$ | social discounter | U-shaped | 72 | 20.28 | 47.61 |
| $d_{i3} > d_{i1} > d_{i2}$ | social magnifier | U-shaped | 51 | 14.37 | 61.97 |
| $d_{i1} = d_{i2} = d_{i3}$ | no discounting | | 33 | 9.30 | 71.27 |
| $d_{i2} > d_{i1} > d_{i3}$ | social discounter | inverse U-shaped | 32 | 9.01 | 80.28 |
| $d_{i3} > d_{i2} > d_{i1}$ | social magnifier | monotone | 27 | 7.61 | 87.89 |
| $d_{i2} > d_{i3} > d_{i1}$ | social magnifier | inverse U-shaped | 19 | 5.35 | 93.24 |
| $d_{i1} > d_{i2} = d_{i3}$ | social discounter | | 11 | 3.10 | 96.34 |
| $d_{i1} = d_{i2} = d_{i3} = 0$ | indifferent in all choices | | 6 | 1.69 | 98.03 |
| $d_{i3} > d_{i1} = d_{i2}$ | social magnifier | | 6 | 1.69 | 99.72 |
| $d_{i2} = d_{i3} > d_{i1}$ | social magnifier | | 1 | 0.28 | 100.00 |
| Total | | | 355 | 100 | |

*Notes.* $d_{i1}$: weight estimated for the most similar passive subject; $d_{i2}$: for the one in the middle who is 50% of the way to the least similar passive subject; $d_{i3}$: for the least similar passive subject.

# Appendix D Extension experiment: additional analyses

## D.1 Choice data

In Section 5.2, we examined the choice data in the original experiment. Here we look at the similar choice data for the extension experiment.

However, in order to make like comparisons, we separate the choice data by the different comparison groups in the extension. Recall that subjects in the extension completed the experiment in two parts. First, for their initial 27 choices, they considered matches at the same social distances as subjects in the original experiment did. Figure D1 reports these results and is, therefore, directly comparable to Figure 4 above. Second, they considered matches at different social distances and took 27 additional choices like before between these three new matches as well as 27 additional choices between self and each of these new matches. Figure D2 reports these results.

In the first series of their choices, we observe a similar pattern of results over the initial 27 choices in Figure D1 as we saw in the original experiment and reported in Figure 4 above. Again, there is not a strict dominance of money nor is there a dominance of the more similar match. Instead, we observe subject making tradeoffs between money and similarity.

In the second series of their choices shown in Figure D2, we include the self as one potential recipient for 27 of the choices and we select matches at different positions in the dissimilarity index. As described in Section 6.1, we included three new matches: the second-most similar passive subject, the passive subject 25% of the way to the least similar match, and the passive subject 75% the way.

We note several observations from Figure D2. First, including self as a potential recipient significantly changes behaviour. Subjects are keen for themselves to receive the money over any other possible match irrespective of the match's similarity. This is especially true in cases where subjects choose between equal payoffs to self and a potential match or a payoff to self which is greater than a payoff to any possible match. Subjects have no qualms about advantageous inequality in these situations. Further, subjects preference for self receiving the payoff remains strong even when such a preference is less efficient than the alternative. The most extreme case is ($1, $5) where choosing self results in the loss of $4 potential dollars. Even in this case, more than 50% of subjects choose to do so which is more than double the percentage of subjects who choose to do so when self is not an option as can be seen in comparison to the lower three bars of the same case in this figure as well as the comparable cases in Figures 4 and D1.

Excluding choices involving self and considering just choices among a subject's three others matches in this second series of choices, we observe a change in the overall pattern of behaviour. Considering the patterns between "Similar, Intermediate" matches, the "Similar, Dissimilar" matches, and the "Similar, Dissimilar" matches, we see in all instances a stronger preference for the more similar match than in the first series of choices of the extension experiment or in the original experiment. We briefly discuss two possible explanations for this. First, the matches here are all relatively closer to the subject, so perhaps what we had previously observed was a preference more the most

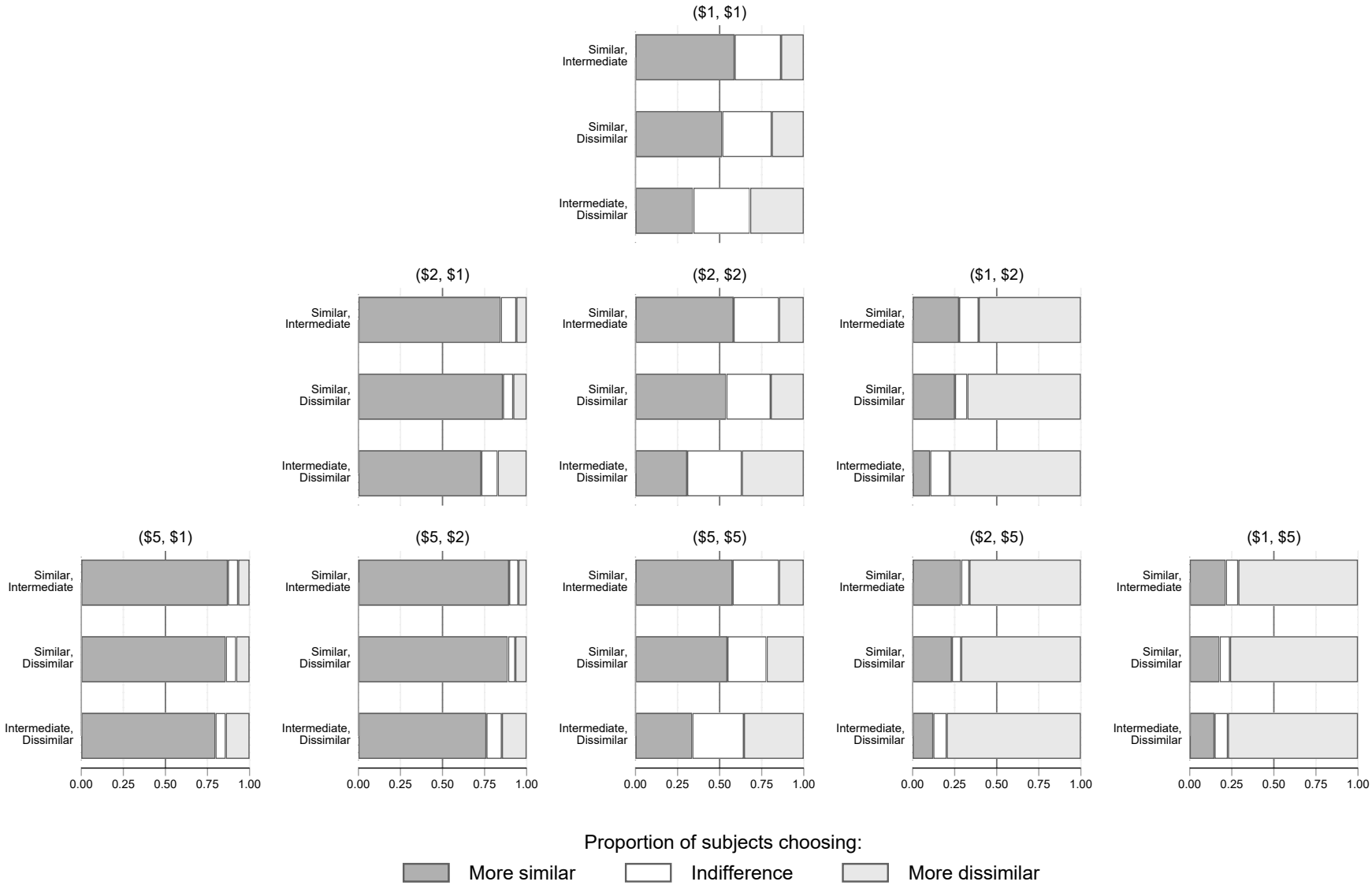



**Figure D1:** Stacked bar graphs showing proportion of choices over monetary amounts and matches, extension experiment, first session excluding self.

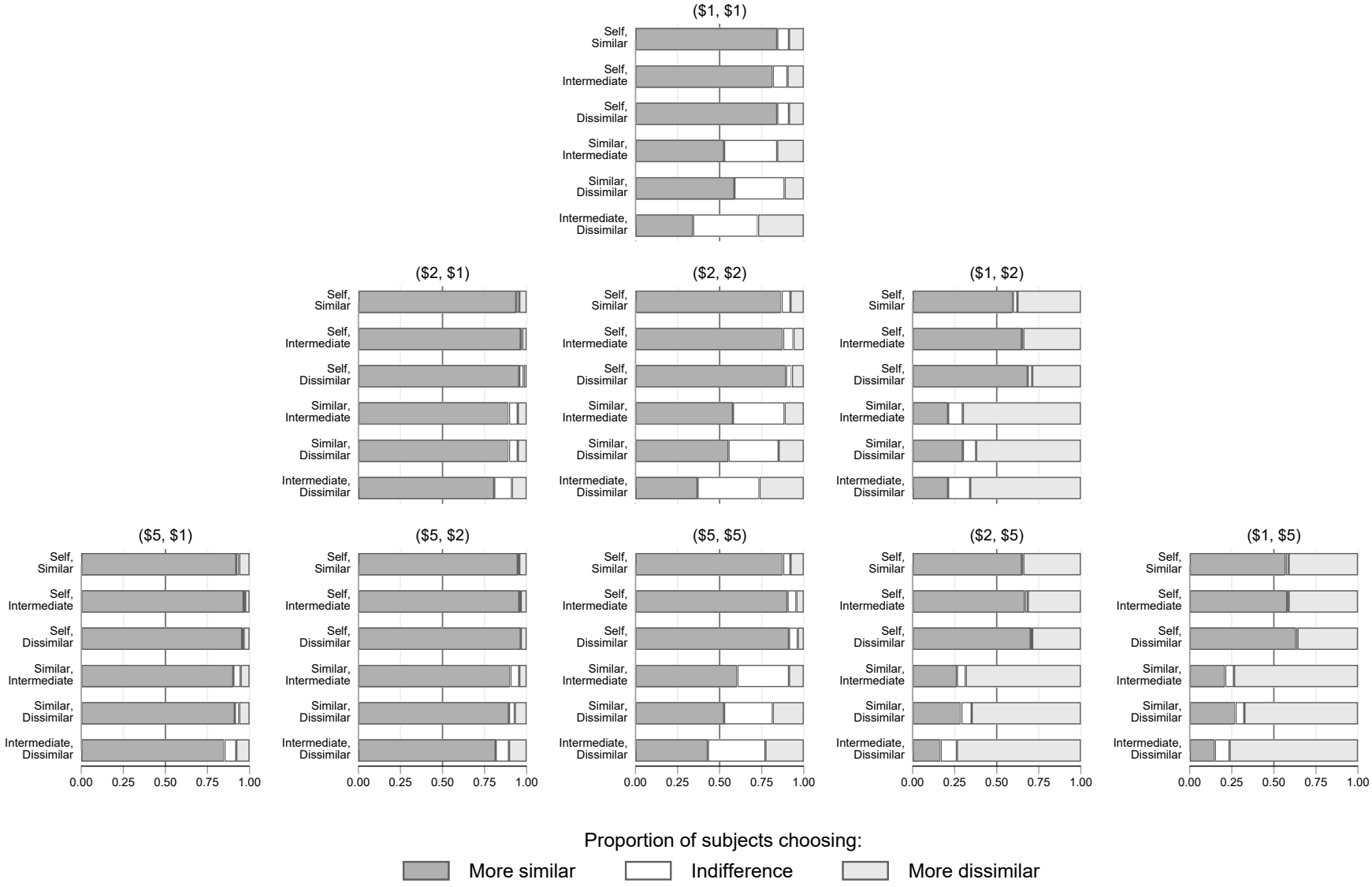



**Figure D2:** Stacked bar graphs showing proportion of choices over monetary amounts and matches, extension experiment, second session including self.

dissimilar subject and that same preference does not hold for the passive 75% the way different. Second, including self in this part of the extension experiment might function as a focal point which causes every other match to seem more dissimilar than when self is not a possibility. In this instance, the "gravitational pull" of self might be so strong that preferences observable when self is excluded might no longer register. This could be an interesting area of inquiry for future studies.

## D.2 Nonparametric estimation

Here we reproduce the analysis of Section 5.3 with the data from the extension experiment. We analyse the two sessions separately: since subjects did not see the matches from the first sessions and those from the second session at the same time, the nonparametric estimation does not allow us to compare the $d_{ij}$ estimated in each session.

### D.2.1 First session: no self included

Table D1 reports the number and the proportion of subjects for each ordering.[33] Figure D3 shows the $d_{ij}$ as a function of the dissimilarity index separately for the six most common orderings, which represent about 92% of subjects.

We find that about 22% of subjects are social magnifiers and about 64% are social discounters.

### D.2.2 Second session: self included

Table D2 reports the number and the proportion of subjects for each ordering.[34] Figure D4 shows the $d_{ij}$ as a function of the dissimilarity index separately for the six most common orderings, which represent about 81% of subjects.

We find that about 15% of subjects are social magnifiers and about 66% are social discounters. In addition, we find that the majority of subjects (about 88%) put more weight on self than on any of the matches.

[33] We can recover estimates via maximum likelihood for 90 out of 109 subjects. Among them, 2 are indifferent in every choice and form a separate category. As in the main text, we recover the ordering of the remaining 19 subjects by revealed preference. These subjects are not shown in Figure D3 since they do not have $d_{ij}$ estimates, but we include them in Table D1.

[34] For this session, we can recover estimates via maximum likelihood for 101 out of 109 subjects. We recover the ordering of the remaining 8 subjects by revealed preference. These subjects are not shown in Figure D4 since they do not have $d_{ij}$ estimates, but we include them in Table D2.

**Table D1:** Orderings resulting from $d_{ij}$ estimated, extension experiment, first session.

| Ordering | Type | Shape | $n$ | % | Cumulative % |
|---|---|---|---|---|---|
| $d_{i1} > d_{i2} > d_{i3}$ | social discounter | monotone | 33 | 30.28 | 30.28 |
| $d_{i1} > d_{i3} > d_{i2}$ | social discounter | U-shaped | 31 | 28.44 | 58.72 |
| $d_{i1} = d_{i2} = d_{i3}$ | no discounting | | 13 | 11.93 | 70.64 |
| $d_{i3} > d_{i1} > d_{i2}$ | social magnifier | U-shaped | 12 | 11.01 | 81.65 |
| $d_{i3} > d_{i2} > d_{i1}$ | social magnifier | monotone | 6 | 5.50 | 87.16 |
| $d_{i2} > d_{i3} > d_{i1}$ | social magnifier | inverse U-shaped | 5 | 4.59 | 91.74 |
| $d_{i2} > d_{i1} > d_{i3}$ | social discounter | inverse U-shaped | 4 | 3.67 | 95.41 |
| $d_{i1} = d_{i2} = d_{i3} = 0$ | indifferent in all choices | | 2 | 1.83 | 97.25 |
| $d_{i1} > d_{i2} = d_{i3}$ | social discounter | | 2 | 1.83 | 99.08 |
| $d_{i3} > d_{i1} = d_{i2}$ | social magnifier | | 1 | 0.92 | 100.00 |
| Total | | | 109 | 100 | |

*Notes.* $d_{i1}$: weight estimated for the most similar passive subject; $d_{i2}$: for the one in the middle who is 50% of the way to the least similar passive subject; $d_{i3}$: for the least similar passive subject.

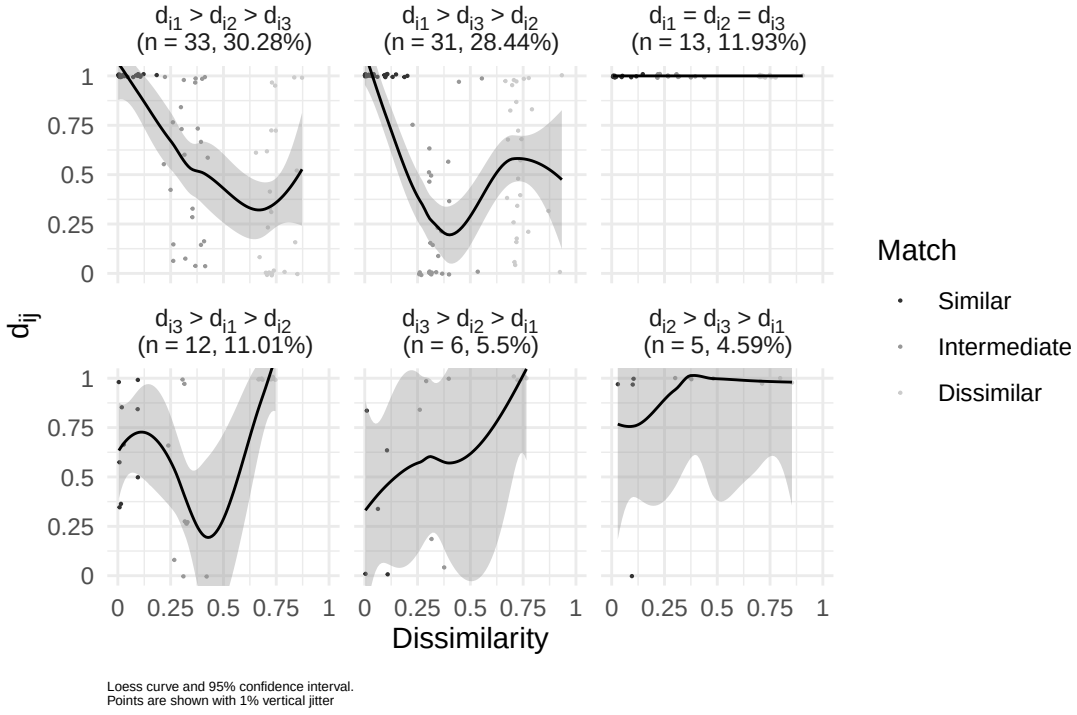



**Figure D3:** Scatterplot of dissimilarity index and $d_{ij}$ estimated, separately for each of the most common orderings. Extension experiment, first session.

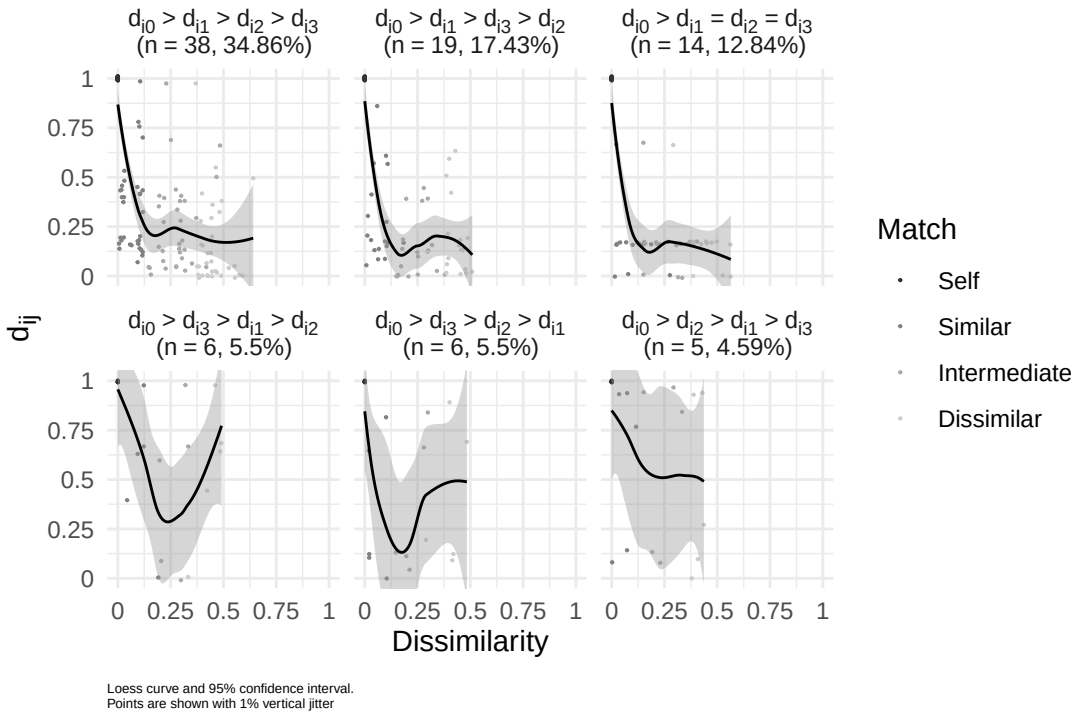



**Figure D4:** Scatterplot of dissimilarity index and $d_{ij}$ estimated, separately for each of the most common orderings. Extension experiment, second session with self included.

**Table D2:** Orderings resulting from $d_{ij}$ estimated, extension experiment, second session with self included.

| Ordering | Bias | Type | Shape | $n$ | % | Cumulative % |
|---|---|---|---|---|---|---|
| $d_{i0} > d_{i1} > d_{i2} > d_{i3}$ | self premium | social discounter | monotone | 38 | 34.86 | 34.86 |
| $d_{i0} > d_{i1} > d_{i3} > d_{i2}$ | self premium | social discounter | U-shaped | 19 | 17.43 | 52.29 |
| $d_{i0} > d_{i1} = d_{i2} = d_{i3}$ | self premium | no discounting | | 14 | 12.84 | 65.14 |
| $d_{i0} > d_{i3} > d_{i1} > d_{i2}$ | self premium | social magnifier | U-shaped | 6 | 5.50 | 70.64 |
| $d_{i0} > d_{i3} > d_{i2} > d_{i1}$ | self premium | social magnifier | monotone | 6 | 5.50 | 76.15 |
| $d_{i0} > d_{i2} > d_{i1} > d_{i3}$ | self premium | social discounter | inverse U-shaped | 5 | 4.59 | 80.73 |
| $d_{i0} = d_{i1} = d_{i2} = d_{i3}$ | | no discounting | | 3 | 2.75 | 83.49 |
| $d_{i0} > d_{i1} = d_{i3} > d_{i2}$ | self premium | | | 2 | 1.83 | 85.32 |
| $d_{i0} > d_{i1} > d_{i2} = d_{i3}$ | self premium | social discounter | | 2 | 1.83 | 87.16 |
| $d_{i0} > d_{i2} > d_{i3} > d_{i1}$ | self premium | social magnifier | inverse U-shaped | 2 | 1.83 | 88.99 |
| $d_{i1} = d_{i2} > d_{i3} > d_{i0}$ | other premium | social discounter | | 2 | 1.83 | 90.83 |
| $d_{i1} > d_{i3} > d_{i2} > d_{i0}$ | other premium | social discounter | U-shaped | 2 | 1.83 | 92.66 |
| $d_{i0} > d_{i1} = d_{i2} > d_{i3}$ | self premium | social discounter | | 1 | 0.92 | 93.58 |
| $d_{i0} > d_{i2} > d_{i1} = d_{i3}$ | self premium | | | 1 | 0.92 | 94.50 |
| $d_{i1} = d_{i2} = d_{i3} > d_{i0}$ | other premium | no discounting | | 1 | 0.92 | 95.41 |
| $d_{i1} > d_{i0} > d_{i2} = d_{i3}$ | other premium | social discounter | | 1 | 0.92 | 96.33 |
| $d_{i1} > d_{i0} > d_{i2} > d_{i3}$ | other premium | social discounter | monotone | 1 | 0.92 | 97.25 |
| $d_{i1} > d_{i2} > d_{i0} > d_{i3}$ | other premium | social discounter | monotone | 1 | 0.92 | 98.17 |
| $d_{i2} > d_{i3} > d_{i1} > d_{i0}$ | other premium | social magnifier | inverse U-shaped | 1 | 0.92 | 99.08 |
| $d_{i3} > d_{i2} > d_{i0} > d_{i1}$ | other premium | social magnifier | monotone | 1 | 0.92 | 100.00 |
| Total | | | | 109 | 100 | |

*Notes.* $d_{i1}$: weight estimated for the second-most similar passive subject; $d_{i2}$: for the one who is 25% of the way to the least similar passive subject; $d_{i3}$: for the one who is 75% of the way to the least similar passive subject.

# Appendix E Preference for more dissimilar matches: Redistributive concerns

Here we provide more information on the redistributive concern analysis reported in Section 7.1.2 of the main text.

Every five years, the American Community Survey generates estimates of median earnings in the past 12 months for the population 16 years and over. There are separate estimates for each zip code in the United States with different measures by sex according to race and ethnicity. There are estimates for seven race categories by sex. The categories of race include White Alone; Black or African American; American Indian and Alaska Native Alone; Asian Alone; Native Hawaiian and Other Pacific Islander Alone; Some Other Race Alone; or Two or More Races. Separately, there are estimates for each ethnicity by sex; that is, there are separate estimates for those who are Hispanic or Latino and those they identify as "White Alone, Not Hispanic or Latino". Not every combination of sex, race, and ethnicity is present in every zip code in the United States at sufficient levels to generate estimates of median earnings. Nevertheless, this American Community Survey data provides a unique way to address the possibility of redistributive concerns influencing the decisions of our subjects.

In order to integrate this income data into our analysis, it is necessary for us to make some choices. In our study, we report both the race and ethnicity of each match to our subjects. Since the American Community Survey generates separate estimates of median earnings for race and ethnicity, we have to decide what to do when there are two separate estimates for a match.

In the main text, when a subject reports being of Hispanic, Latino, or Spanish origin, we report their estimate by ethnicity regardless of race. When a subject reports not being of Hispanic, Latino, or Spanish origin, we report their estimate by race. There is one exception to this rule which occurs for subjects who consider their race to be white and report not being of Hispanic, Latino, or Spanish origin. In this instance, we could either select the "White Alone" estimate by race or the "White Alone, Not Hispanic or Latino" estimate by ethnicity. To resolve this discrepancy, we use the "White Alone, Not Hispanic or Latino" estimate by ethnicity.

We also carry out robustness checks with three other ways of reconciling the discrepancy between the ACS estimates. First, we average the estimates of median earnings by their racial category and the Hispanic and Latino category in a given zip code by sex. Second, we take the minimum estimate of median earnings between their racial category and the Hispanic or Latino category. The third approach was to always choose the maximum of the race and ethnicity median earnings for those reporting their ethnicity to be Hispanic or Latino.

We find the same results under these three alternative assumptions. We also find the same results when we carry out the analysis using the data from the extension experiment or the pooled data.

# Appendix F Pooled data analysis: alternative functional forms and distributional assumptions

## F.1 Data

In the main text we estimated the structural model twice: once with the original dataset (Section 5.4.3) and a second time with the extension dataset (Section 6). Here we do the same estimations but with all the data pooled. We are also able to include subjects we were unable to include in the main text: those who only completed the first session of the extension experiment.

## F.2 Utility of money $u(x)$

For the utility of money—$u(x)$—we use two functional forms:

- Exponential:

$$u(x, \alpha_i) = \begin{cases} 1 - \exp(-\alpha_i x) & \text{if } \alpha_i > 0, \\ x & \text{if } \alpha_i = 0, \\ \exp(-\alpha_i x) - 1 & \text{if } \alpha_i < 0, \end{cases} \tag{4}$$

  where $u(x)$ is concave for $\alpha_i > 0$, convex for $\alpha_i < 0$, and linear for $\alpha_i = 0$;
- Power:

$$u(x, \alpha_i) = \begin{cases} x^{\alpha_i} & \text{if } \alpha_i > 0, \\ \log(x) & \text{if } \alpha_i = 0, \\ -x^{\alpha_i} & \text{if } \alpha_i < 0, \end{cases} \tag{5}$$

  where $u(x)$ is concave for $\alpha_i < 1$ and convex for $\alpha_i > 1$, and linear for $\alpha_i = 1$.

We need to make an additional assumption in order to use of the power utility function. Using the same notation as in the main text, in choice $t$ the left option $\omega_t^L$ gives payoff $x_t^L$ to a recipient at dissimilarity $s_t^L$ and the right option $\omega_t^R$ gives payoff $x_t^R$ to a recipient at dissimilarity $s_t^R$. In both cases, the other recipient receives nothing.

Dropping the subscript $i$ for readability, the utility of the left option is therefore given by

$$U(\omega_t^L) = D(s_t^L)\, u(x_t^L) + D(s_t^R)\, u(0),$$

and similarly for the right option. The difference in utility is

$$\Delta U_t = D(s_t^L)\, u(x_t^L) - D(s_t^R)\, u(x_t^R) + u(0)\Big[D(s_t^R) - D(s_t^L)\Big]. \tag{6}$$

The last term shows that, unlike in expected utility, the level of $u(x)$ is not a free normalisation here. Adding a constant to $u(x)$ changes the predicted choices because $u(x)$ is multiplied by $D(s)$, which differs across the two recipients.

When using the exponential utility function, $u(0) = 0$ for every $\alpha_i$ and the last term of (6) vanishes. However, when using the power utility function, this is no longer true. For $\alpha_i < 0$, utility is negative for every amount and only approaches 0 as the amount goes to infinity. If we drop the last term of (6), the meaning of $D(s)$ is reversed. In this case, a lower $D(s)$ shrinks a negative utility, so a subject with $\delta < 1$ would favour the more dissimilar recipient.

Restricting $\alpha_i > 0$ might seem like an attractive solution, but as argued by Wakker (2008), it is not. Doing so leaves out the most concave part of the power family, which might be needed to achieve a good fit. This is a known issue with the power utility function.

We follow the advice of Wakker (2008), as do Andersen et al. (2008) and Andreoni and Sprenger (2012), and add a level of background wealth $w > 0$ to the amounts of money. The utility of the left option becomes

$$U(\omega_t^L) = D(s_t^L)\, u(w + x_t^L) + D(s_t^R)\, u(w),$$

and the right option is similarly modified. The difference in utility becomes

$$\Delta U_t = D(s_t^L)\Big[u(w + x_t^L) - u(w)\Big] - D(s_t^R)\Big[u(w + x_t^R) - u(w)\Big]. \tag{7}$$

This modification sidesteps the issue because $u(w)$ is finite for every $\alpha_i$ when $w > 0$.

Thus, the recipient who receives nothing can be kept in the model, and each recipient's utility is measured as the gain $v(x) = u(w + x) - u(w)$ over what they would have without the transfer. This gain is zero when $x = 0$, positive when $x > 0$, and increasing in $x$ for every $\alpha_i$, including $\alpha_i \leq 0$. The social discount function $D(s)$ therefore keeps its meaning whatever the curvature of $u(x)$. The parameter $\alpha_i$ remains unrestricted, and $\alpha_i < 0$ simply means a utility function more concave than the logarithmic function.

Background wealth is not a neutral normalisation, however: its level affects the estimated parameters (Andreoni and Sprenger, 2012). Conceptually, $w$ should be the wealth, in the subject's mind, with which the recipient integrates the payoff. The natural candidate is what subjects think the recipients earned for taking part in the survey. Subjects knew that the recipients had "completed the same survey as you" and were "selected from several hundred MTurk Workers who previously completed the same survey as you" (Appendix A.1.1 and A.1.3). Therefore, subjects could reasonably infer that the recipients, like themselves, had been paid for completing the survey.

Subjects who inferred this would have been correct as we had paid the survey participants \$1.25 each. However, we did not explicitly tell this to the subjects in this experiment. Nonetheless, subjects could reasonably estimate this amount, since they had completed the same survey themselves. Further, subjects knew the fixed rate we were paying them for completing the experiment—\$3.00—and had a rough estimate of the time the entire experiment would last—about 20 minutes (median). Having completed the survey themselves, they could recall that it took them 6.5 minutes (median). Scaling the \$3.00 they were to receive for about 20 minutes (median) of work to the length of the survey, they could reasonably assume that their matches had received about \$1.00. Alternatively, subjects could have assumed that we paid those who completed the survey

the federal minimum wage. At the federal minimum wage of \$7.25 per hour, 6.5 minutes of work is worth about \$0.80.

For these reasons, we run the estimations with $w = 0.5$, 1, and 1.5. We report all results below. We find that $w = 1$ provides the overall best fit, which is why we report the model with this as the background wealth in the main text.

## F.3 Social discount function $D(s)$ and distribution of $\delta$

Social discount functions that do not allow for self bias—no discounting, exponential, and quadratic-exponential—are poorly identified on this pooled sample. This is because the pooled sample includes the extension data in which subjects trade off money for themselves against money for others. Subjects' strong preference for themselves is absorbed into a degenerate estimate of $\delta$ rather than into a separate self-bias parameter in these social discount functions. We saw this in the analysis of the extension experiment (Section 6, and especially Figure 8). Therefore, we use the two functional forms that include self bias—quasi-hyperbolic and quasi-hyperbolic quadratic discounting—in this appendix.

For $D(s)$ we use:

- Quasi-hyperbolic:

$$D(s) = \begin{cases} 1 & \text{if } s = 0, \\ \beta\,\delta^{s} & \text{if } s > 0, \end{cases} \tag{8}$$

- Quasi-hyperbolic quadratic:

$$D(s) = \begin{cases} 1 & \text{if } s = 0, \\ \beta\,\delta^{s+\kappa s^2} & \text{if } s > 0, \end{cases} \tag{9}$$

where $s = 0$ represents the self, $s > 0$ represents others, and $\beta > 0$ captures self bias.

For the distribution of $\delta$ we use:

- Lognormal, $\ln \delta \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2)$, and
- Normal, $\delta \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2)$, left-truncated at 0.

## F.4 Estimation results

The following tables and figures report the results:

- Tables F1–F4 report the maximum simulated likelihood estimates, with one table per utility function. Table F1 reports the models with $u(x)$ exponential. Tables F2, F3 and F4 report the models with $u(x)$ power and background wealth $w = 0.5$, 1 and 1.5. Each table contains the four combinations of $D(s)$ and $\delta$ distribution, for a total of 16 models.
- Figure F1 plots the estimated distributions of $\alpha$, $\beta$, $\delta$ and $\kappa$ for these models.
- Tables F5 and F6 report the moments of the estimated distributions of $\delta$ and $\beta$ and hypothesis tests.[35]

[35]For the lognormal distribution, the moments are computed as in the main text as mean($\delta$) =

- Table F7 reports the corresponding proportion of social magnifiers for each specification.
- Finally, Table F8 assesses the fit of the models and compares them one to another.

As we can see, the model with $u(x)$ power and background wealth $w = 1$ provides the best fit compared to those with $w = 0.5$ and $w = 1.5$, which is why we report the model with $w = 1$ in the main text. It even provides a slightly better fit in terms of log-likelihood than the model with $u(x)$ exponential, for a similar hit rate. However, it requires an extra assumption on the level of background wealth $w$, which is why we have focused on the model with $u(x)$ exponential in the main text. In any case, the estimates of the parameters of interest are similar across the models.

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$\exp\left(\mu_\delta + \frac{\sigma_\delta^2}{2}\right)$, $\text{median}(\delta) = \exp(\mu_\delta)$, and $\text{mode}(\delta) = \exp\left(\mu_\delta - \sigma_\delta^2\right)$. For the truncated normal distribution, the moments should in principle be computed as $\text{mean}(\delta) = \mu_\delta + \sigma_\delta \frac{\phi\left(\frac{-\mu_\delta}{\sigma_\delta}\right)}{1-\Phi\left(\frac{-\mu_\delta}{\sigma_\delta}\right)}$, $\text{median}(\delta) = \mu_\delta + \sigma_\delta\, \Phi^{-1}\left(\frac{\Phi\left(\frac{-\mu_\delta}{\sigma_\delta}\right)+1}{2}\right)$, and $\text{mode}(\delta) = \max(0, \mu_\delta)$, where $\phi(\cdot)$ is the probability density function of the standard normal distribution, $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution, and $\Phi^{-1}(\cdot)$ is its inverse. However, the software implementation used to apply the delta method requires explicit symbolic differentiation and does not natively support $\Phi(\cdot)$. To bypass this software limitation, we use the approximation of $\Phi(\cdot)$ offered by Lipoth et al. (2022) who show it is the one that results in the smallest approximation error.

**Table F1:** Maximum simulated likelihood estimates, pooled data, $u(x)$ exponential.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| $D(s)$ | Quasi-hyp. | Quasi-hyp. | Quasi-hyp. quad. | Quasi-hyp. quad. |
| $\delta$ distribution | Lognormal | Truncated | Lognormal | Truncated |
| $\mu_\alpha$ | $1.31^{***}$ | $0.53^{***}$ | $1.86^{***}$ | $1.91^{***}$ |
| | (0.09) | (0.04) | (0.09) | (0.14) |
| $\sigma_\alpha$ | $1.36^{***}$ | $0.52^{***}$ | $1.64^{***}$ | $1.69^{***}$ |
| | (0.09) | (0.04) | (0.08) | (0.13) |
| $\mu_\delta$ | $-0.06^{***}$ | $0.62^{***}$ | $-0.36^{***}$ | $0.70^{***}$ |
| | (0.01) | (0.06) | (0.03) | (0.09) |
| $\sigma_\delta$ | $0.26^{***}$ | $0.61^{***}$ | $0.59^{***}$ | $0.46^{***}$ |
| | (0.03) | (0.05) | (0.04) | (0.10) |
| $\mu_\beta$ | $-0.44^{***}$ | $-0.62^{***}$ | $-0.18^{***}$ | $-0.16^{***}$ |
| | (0.05) | (0.07) | (0.02) | (0.03) |
| $\sigma_\beta$ | $0.74^{***}$ | $1.29^{***}$ | $0.39^{***}$ | $0.36^{***}$ |
| | (0.07) | (0.06) | (0.03) | (0.04) |
| $\mu_\kappa$ | | | $-1.03^{***}$ | $-1.01^{***}$ |
| | | | (0.00) | (0.01) |
| $\sigma_\kappa$ | | | $0.40^{***}$ | $0.41^{***}$ |
| | | | (0.02) | (0.01) |
| $\sigma$ | $0.07^{***}$ | $0.07^{***}$ | $0.07^{***}$ | $0.07^{***}$ |
| | (0.00) | (0.01) | (0.00) | (0.01) |
| Log Likelihood | −8600.51 | −8569.38 | −7951.96 | −7949.13 |

*Notes.* Robust standard errors in parentheses.
$u(x)$ exponential as defined in (4).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\alpha$ and $\kappa$ normally distributed, $\beta$ log-normally distributed, $\delta$ log-normally distributed or normally distributed and left-truncated at 0.
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

**Table F2:** Maximum simulated likelihood estimates, pooled data, $u(x)$ power, $w = 0.5$.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| $D(s)$ | Quasi-hyp. | Quasi-hyp. | Quasi-hyp. quad. | Quasi-hyp. quad. |
| $\delta$ distribution | Lognormal | Truncated | Lognormal | Truncated |
| $\mu_\alpha$ | $-0.14^{***}$ | $0.09^{***}$ | $-0.34^{***}$ | $-0.29$ |
| | (0.00) | (0.00) | (0.07) | (0.17) |
| $\sigma_\alpha$ | $1.28^{***}$ | $0.37^{***}$ | $1.51^{***}$ | $1.35^{***}$ |
| | (0.09) | (0.00) | (0.07) | (0.12) |
| $\mu_\delta$ | $-0.25^{***}$ | $-0.04$ | $-0.56^{***}$ | $0.13$ |
| | (0.03) | (4.99) | (0.08) | (0.69) |
| $\sigma_\delta$ | $0.42^{***}$ | $0.80$ | $1.04^{***}$ | $0.83^{*}$ |
| | (0.04) | (2.05) | (0.14) | (0.33) |
| $\mu_\beta$ | $-1.04^{***}$ | $-0.77^{***}$ | $-0.83^{***}$ | $-0.81^{***}$ |
| | (0.07) | (0.04) | (0.06) | (0.13) |
| $\sigma_\beta$ | $0.72^{***}$ | $0.88^{***}$ | $0.60^{***}$ | $0.60^{***}$ |
| | (0.05) | (0.05) | (0.04) | (0.09) |
| $\mu_\kappa$ | | | $-1.12^{***}$ | $-0.89^{***}$ |
| | | | (0.01) | (0.04) |
| $\sigma_\kappa$ | | | $0.43^{***}$ | $0.52^{***}$ |
| | | | (0.02) | (0.04) |
| $\sigma$ | $0.14^{***}$ | $0.05^{***}$ | $0.14^{***}$ | $0.12^{***}$ |
| | (0.02) | (0.01) | (0.01) | (0.01) |
| Log Likelihood | $-8541.23$ | $-8464.43$ | $-7963.46$ | $-7947.77$ |

*Notes.* Robust standard errors in parentheses.
$u(x)$ power as defined in (5), with background wealth $w = 0.5$ as in (7).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\alpha$ and $\kappa$ normally distributed, $\beta$ log-normally distributed, $\delta$ log-normally distributed or normally distributed and left-truncated at 0.
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

**Table F3:** Maximum simulated likelihood estimates, pooled data, $u(x)$ power, $w = 1$.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| $D(s)$ | Quasi-hyp. | Quasi-hyp. | Quasi-hyp. quad. | Quasi-hyp. quad. |
| $\delta$ distribution | Lognormal | Truncated | Lognormal | Truncated |
| $\mu_\alpha$ | $-1.97^{***}$ | $-1.84^{***}$ | $-2.63^{***}$ | $-2.08^{***}$ |
| | (0.21) | (0.07) | (0.12) | (0.17) |
| $\sigma_\alpha$ | $2.56^{***}$ | $2.41^{***}$ | $2.74^{***}$ | $2.71^{***}$ |
| | (0.27) | (0.09) | (0.12) | (0.22) |
| $\mu_\delta$ | $-0.08^{***}$ | $0.92^{***}$ | $-0.37^{***}$ | 0.39 |
| | (0.01) | (0.01) | (0.03) | (0.23) |
| $\sigma_\delta$ | $0.17^{***}$ | $0.21^{***}$ | $0.48^{***}$ | $0.67^{***}$ |
| | (0.03) | (0.02) | (0.04) | (0.14) |
| $\mu_\beta$ | $-0.43^{***}$ | $-0.57^{***}$ | $-0.22^{***}$ | $-0.32^{***}$ |
| | (0.09) | (0.02) | (0.02) | (0.04) |
| $\sigma_\beta$ | $0.58^{***}$ | $0.79^{***}$ | $0.27^{***}$ | $0.27^{***}$ |
| | (0.11) | (0.03) | (0.02) | (0.03) |
| $\mu_\kappa$ | | | $-0.98^{***}$ | $-1.10^{***}$ |
| | | | (0.01) | (0.01) |
| $\sigma_\kappa$ | | | $0.38^{***}$ | $0.41^{***}$ |
| | | | (0.02) | (0.02) |
| $\sigma$ | $0.06^{***}$ | $0.06^{***}$ | $0.05^{***}$ | $0.05^{***}$ |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| Log Likelihood | −8474.43 | −8473.54 | −7834.07 | −7841.90 |

*Notes.* Robust standard errors in parentheses.
$u(x)$ power as defined in (5), with background wealth $w = 1$ as in (7).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\alpha$ and $\kappa$ normally distributed, $\beta$ log-normally distributed, $\delta$ log-normally distributed or normally distributed and left-truncated at 0.
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

**Table F4:** Maximum simulated likelihood estimates, pooled data, $u(x)$ power, $w = 1.5$.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| $D(s)$ | Quasi-hyp. | Quasi-hyp. | Quasi-hyp. quad. | Quasi-hyp. quad. |
| $\delta$ distribution | Lognormal | Truncated | Lognormal | Truncated |
| $\mu_\alpha$ | $-0.96^{***}$ | $-2.07^{***}$ | $-1.72^{***}$ | $-1.59^{***}$ |
| | (0.05) | (0.14) | (0.07) | (0.14) |
| $\sigma_\alpha$ | $2.10^{***}$ | $2.14^{***}$ | $1.85^{***}$ | $2.41^{***}$ |
| | (0.10) | (0.15) | (0.08) | (0.20) |
| $\mu_\delta$ | $-0.20^{***}$ | $0.87^{***}$ | $-0.88^{***}$ | $-4.96^{***}$ |
| | (0.01) | (0.01) | (0.07) | (0.59) |
| $\sigma_\delta$ | $0.50^{***}$ | $0.37^{***}$ | $1.32^{***}$ | $1.90^{***}$ |
| | (0.02) | (0.03) | (0.11) | (0.10) |
| $\mu_\beta$ | $-0.72^{***}$ | $-0.36^{***}$ | $-0.50^{***}$ | $-0.26^{***}$ |
| | (0.02) | (0.07) | (0.02) | (0.03) |
| $\sigma_\beta$ | $0.94^{***}$ | $0.45^{***}$ | $0.33^{***}$ | $0.45^{***}$ |
| | (0.03) | (0.04) | (0.03) | (0.04) |
| $\mu_\kappa$ | | | $-1.04^{***}$ | $-1.02^{***}$ |
| | | | (0.01) | (0.01) |
| $\sigma_\kappa$ | | | $0.40^{***}$ | $0.55^{***}$ |
| | | | (0.03) | (0.01) |
| $\sigma$ | $0.04^{***}$ | $0.03^{***}$ | $0.03^{***}$ | $0.03^{***}$ |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| Log Likelihood | −8573.00 | −8434.44 | −7958.95 | −7957.66 |

*Notes.* Robust standard errors in parentheses.
$u(x)$ power as defined in (5), with background wealth $w = 1.5$ as in (7).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\alpha$ and $\kappa$ normally distributed, $\beta$ log-normally distributed, $\delta$ log-normally distributed or normally distributed and left-truncated at 0.
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$.

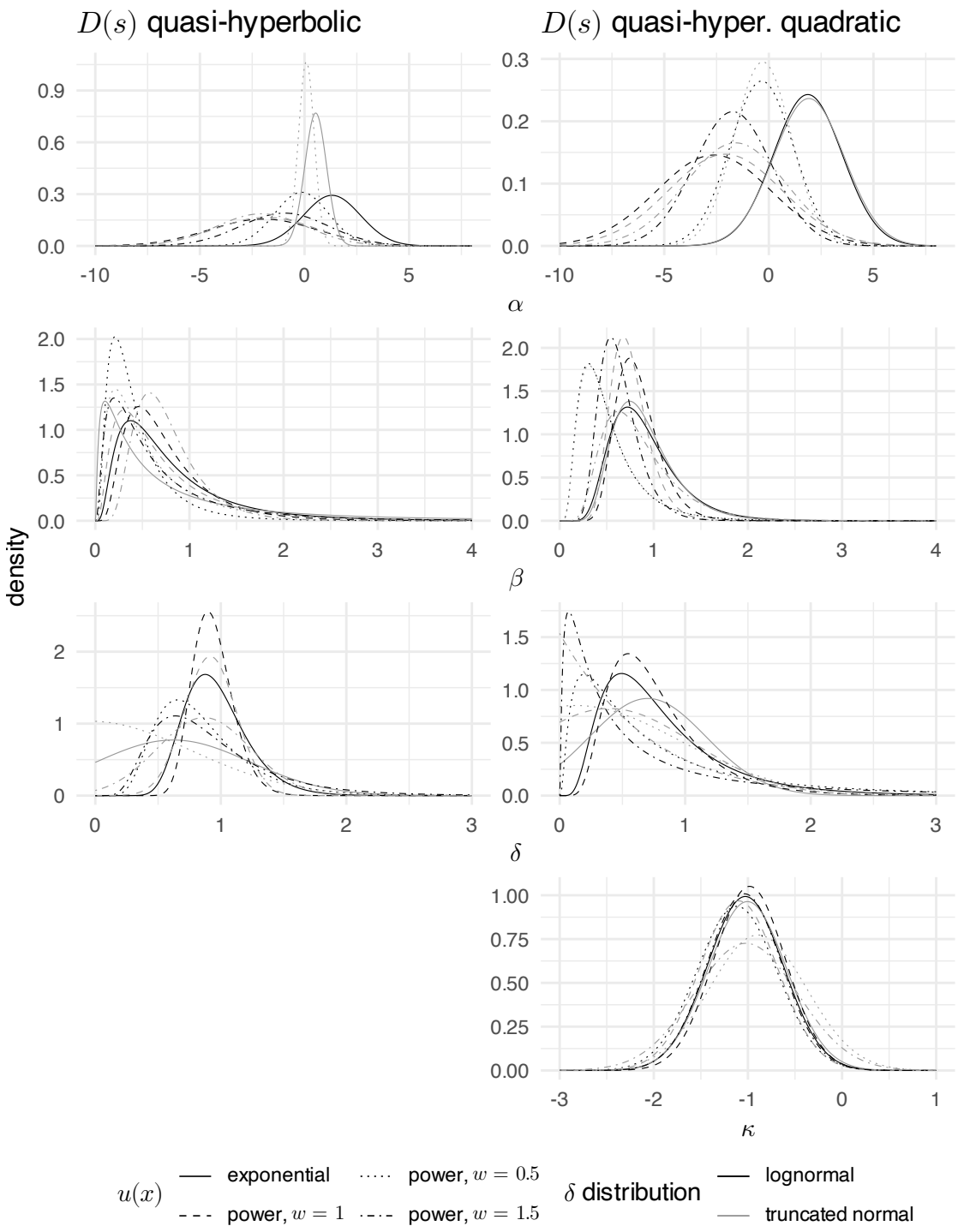



**Figure F1:** Plots of the distributions of $\alpha$, $\beta$, $\delta$ and $\kappa$ estimated in Tables F1–F4.

**Table F5:** Moments of the distribution of $\delta$ and hypothesis tests for the models estimated in Tables F1–F4.

| $u(x)$ | $D(s)$ | $\delta$ distribution | Moment | | Test against 1 | |
|---|---|---|---|---|---|---|
| | | | | | $\chi^2$ | $p$ |
| Exponential | Quasi-hyperbolic | Lognormal | mean | 0.97 | 5832.09 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Lognormal | median | 0.94 | 83.55 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Lognormal | mode | 0.88 | 39.10 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Truncated normal | mean | 0.79 | 5696.64 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Truncated normal | median | 0.74 | 324.23 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Truncated normal | mode | 0.62 | 39.62 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | mean | 0.83 | 38215.38 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | median | 0.70 | 252.32 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | mode | 0.49 | 165.80 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.77 | 99.93 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | median | 0.74 | 34.14 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.70 | 10.58 | 0.001 |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | mean | 0.85 | 243.70 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | median | 0.78 | 101.21 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | mode | 0.66 | 66.84 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | mean | 0.63 | 9.20 | 0.002 |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | median | 0.53 | 1.33 | 0.249 |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | mode | 0.00 | $1.35\times10^{10}$ | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | mean | 0.98 | 0.07 | 0.795 |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | median | 0.57 | 95.22 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | mode | 0.20 | 123.81 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.71 | 428.33 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | median | 0.61 | 36.81 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.13 | 1.60 | 0.206 |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | mean | 0.94 | 50.38 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | median | 0.92 | 31.25 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | mode | 0.90 | 20.22 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | mean | 0.92 | 74.02 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | median | 0.92 | 73.91 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | mode | 0.92 | 73.86 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | mean | 0.78 | 341.21 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | median | 0.69 | 179.27 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | mode | 0.55 | 133.22 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.71 | 870.18 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | median | 0.63 | 88.80 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.39 | 6.76 | 0.009 |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | mean | 0.93 | 262.40 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | median | 0.82 | 406.39 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | mode | 0.64 | 263.36 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | mean | 0.88 | 325.06 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | median | 0.87 | 170.49 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | mode | 0.87 | 101.67 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | mean | 1.00 | 0.00 | 0.978 |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | median | 0.42 | 354.12 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | mode | 0.07 | 1200.69 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.60 | 13931.93 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | median | 0.44 | 19226.91 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.00 | $2.80\times10^{20}$ | $< 0.001$ |

*Notes.* Tests are two-sided Wald tests based on robust standard errors.
$u(x)$ exponential as defined in (4), $u(x)$ power with background wealth $w$ as defined in (5) and (7).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\delta$ truncated normal distribution is left-truncated at 0.

**Table F6:** Moments of the distribution of $\beta$ and hypothesis tests for the models estimated in Tables F1–F4.

| $u(x)$ | $D(s)$ | $\delta$ distribution | Moment | | Test against 1 | |
|---|---|---|---|---|---|---|
| | | | | | $\chi^2$ | $p$ |
| Exponential | Quasi-hyperbolic | Lognormal | mean | 0.85 | 98.77 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Lognormal | median | 0.64 | 135.59 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Lognormal | mode | 0.37 | 120.71 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Truncated normal | mean | 1.24 | 10.69 | 0.001 |
| Exponential | Quasi-hyperbolic | Truncated normal | median | 0.54 | 147.31 | $< 0.001$ |
| Exponential | Quasi-hyperbolic | Truncated normal | mode | 0.10 | 1623.06 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | mean | 0.91 | 168.26 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | median | 0.84 | 138.94 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | mode | 0.72 | 91.81 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.91 | 50.31 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | median | 0.85 | 41.38 | $< 0.001$ |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.74 | 35.22 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | mean | 0.46 | 1121.04 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | median | 0.35 | 718.86 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | mode | 0.21 | 773.94 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | mean | 0.68 | 241.87 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | median | 0.46 | 706.80 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | mode | 0.21 | 808.78 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | mean | 0.52 | 747.89 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | median | 0.44 | 529.76 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | mode | 0.31 | 494.49 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.53 | 147.64 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | median | 0.44 | 99.80 | $< 0.001$ |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.31 | 92.46 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | mean | 0.77 | 133.49 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | median | 0.65 | 40.05 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | mode | 0.46 | 30.35 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | mean | 0.77 | 65.09 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | median | 0.57 | 1000.00 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | mode | 0.30 | 2105.64 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | mean | 0.83 | 147.23 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | median | 0.80 | 172.13 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | mode | 0.74 | 182.16 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.75 | 115.55 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | median | 0.73 | 100.50 | $< 0.001$ |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.68 | 84.34 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | mean | 0.76 | 150.37 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | median | 0.49 | 1874.89 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | mode | 0.20 | 3244.37 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | mean | 0.77 | 21.25 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | median | 0.70 | 41.83 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | mode | 0.57 | 77.93 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | mean | 0.64 | 723.42 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | median | 0.61 | 716.58 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | mode | 0.54 | 524.38 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | mean | 0.85 | 55.89 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | median | 0.77 | 94.95 | $< 0.001$ |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | mode | 0.62 | 91.35 | $< 0.001$ |

*Notes.* Tests are two-sided Wald tests based on robust standard errors.
$u(x)$ exponential as defined in (4), $u(x)$ power with background wealth $w$ as defined in (5) and (7).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\delta$ truncated normal distribution is left-truncated at 0.

**Table F7:** Proportion of social magnifiers, pooled data.

| $u(x)$ | $D(s)$ | $\delta$ distribution | Proportion of magnifiers |
|---|---|---|---|
| Exponential | Quasi-hyperbolic | Lognormal | 0.40 |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | 0.38 |
| Exponential | Quasi-hyperbolic | Truncated normal | 0.32 |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | 0.38 |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | 0.28 |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | 0.42 |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | 0.20 |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | 0.36 |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | 0.32 |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | 0.33 |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | 0.34 |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | 0.39 |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | 0.34 |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | 0.37 |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | 0.36 |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | 0.36 |

*Notes.* For the quasi-hyperbolic social discount function, the proportion of magnifiers is the proportion of the estimated distribution of $\delta$ above 1. For the quasi-hyperbolic quadratic social discount function, this proportion is computed as described in footnote 21 in the main text.
$u(x)$ exponential as defined in (4), $u(x)$ power with background wealth $w$ as defined in (5) and (7).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\delta$ truncated normal distribution is left-truncated at 0.

**Table F8:** Comparison of the fit of the models estimated in Tables F1–F4.

| $u(x)$ | $D(s)$ | $\delta$ distribution | Log-likelihood | Likelihood-ratio test | | Hit rate |
|---|---|---|---|---|---|---|
| | | | | $\chi^2$ | $p$ | |
| Exponential | Quasi-hyperbolic | Lognormal | −8600.51 | | | 0.75 |
| Exponential | Quasi-hyperbolic quadratic | Lognormal | −7951.96 | 1297.10 | $< 0.001$ | 0.78 |
| Exponential | Quasi-hyperbolic | Truncated normal | −8569.38 | | | 0.74 |
| Exponential | Quasi-hyperbolic quadratic | Truncated normal | −7949.13 | 1240.51 | $< 0.001$ | 0.78 |
| Power, $w = 0.5$ | Quasi-hyperbolic | Lognormal | −8541.23 | | | 0.74 |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Lognormal | −7963.46 | 1155.54 | $< 0.001$ | 0.77 |
| Power, $w = 0.5$ | Quasi-hyperbolic | Truncated normal | −8464.43 | | | 0.73 |
| Power, $w = 0.5$ | Quasi-hyperbolic quadratic | Truncated normal | −7947.77 | 1033.32 | $< 0.001$ | 0.77 |
| Power, $w = 1$ | Quasi-hyperbolic | Lognormal | −8474.43 | | | 0.75 |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Lognormal | −7834.07 | 1280.72 | $< 0.001$ | 0.78 |
| Power, $w = 1$ | Quasi-hyperbolic | Truncated normal | −8473.54 | | | 0.75 |
| Power, $w = 1$ | Quasi-hyperbolic quadratic | Truncated normal | −7841.90 | 1263.27 | $< 0.001$ | 0.78 |
| Power, $w = 1.5$ | Quasi-hyperbolic | Lognormal | −8573.00 | | | 0.74 |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Lognormal | −7958.95 | 1228.10 | $< 0.001$ | 0.78 |
| Power, $w = 1.5$ | Quasi-hyperbolic | Truncated normal | −8434.44 | | | 0.75 |
| Power, $w = 1.5$ | Quasi-hyperbolic quadratic | Truncated normal | −7957.66 | 953.56 | $< 0.001$ | 0.78 |

*Notes.* The likelihood-ratio test compares the quasi-hyperbolic quadratic model to the quasi-hyperbolic model with the same $u(x)$ and $\delta$ distribution. Hit rate is the proportion of choices predicted correctly by the model.
$u(x)$ exponential as defined in (4), $u(x)$ power with background wealth $w$ as defined in (5) and (7).
$D(s)$ quasi-hyperbolic as defined in (8), $D(s)$ quasi-hyperbolic quadratic as defined in (9).
$\delta$ truncated normal distribution is left-truncated at 0.

# Appendix G Robson (2021): nonparametric estimation

In this Section we apply a nonparametric version of the parametric estimation described in Section 7.3. The only difference is that, instead of estimating the parameters of $D(s)$, we directly estimate a weight $d_{ij}$ for each of subject $i$'s six possible recipients $j$, one of which is the self, independently for each subject via maximum likelihood, as we did in Section 5.3. We use the same power utility function (3) and the same Dirichlet distribution to model the observed allocations as in Section 7.3, except that here the Dirichlet precision $\varsigma$ is estimated separately for each subject rather than being a single parameter common to all subjects.

Taking this specification to the data requires two further steps. First, the optimal allocation $x^*_{ij}$, and hence the likelihood, is unchanged if every $d_{ij}$ is multiplied by the same positive constant: only the ratios of the six $d_{ij}$ are identified, not their levels. We resolve this by fixing $d_{i0} = 1$ for the self recipient ($j = 0$) and estimating the remaining five weights relative to this anchor. As in Section 5.3, for ease of comparison across subjects we then normalise each subject's six weights by their maximum, $d_{ij}/\max_j d_{ij}$.

Second, the demographic characteristics needed to compute the dissimilarity index $s_{ij}$ are missing for some subjects, so $s_{ij}$ cannot be computed for one or more of the five recipients. Because the ordering we use below labels recipients by their dissimilarity rank—most similar, intermediate, most dissimilar—this labelling is only meaningful for a subject once we know where each of their recipients falls on the dissimilarity scale. We therefore restrict the ordering comparison to subjects for whom $s_{ij}$ can be computed for all five recipients. We can recover valid maximum likelihood estimates for 127 subjects out of 149; of these, 91 have complete demographic information for all five recipients, and Table G1 is based on this sample of 91.

Third, since we have six $d_{ij}$ for each subject (one for the self plus five others), we have 720 possible orderings of the $d_{ij}$ values. We never observe more than a few subjects with the same ordering, which renders difficult to draw conclusions about the most common orderings. Instead, we look at the orderings based only on $d_{i0}$ (self), $d_{i1}$ (most similar recipient), $d_{i3}$ (intermediate recipient) and $d_{i5}$ (most dissimilar recipient).

Table G1 reports the results. As in Section 5.3, we classify subjects with $d_{i1} > d_{i5}$ as social discounters and subjects with $d_{i5} > d_{i1}$ as social magnifiers, irrespective of the position of the intermediate recipient $d_{i3}$. As in the parametric estimation (Section 7.3), we find that self bias and social magnification are both common: 89% of subjects place the largest weight on themselves, and, looking only at the two extremes of the dissimilarity scale, 55% of subjects are social discounters against 44% who are social magnifiers. As in

the parametric estimation, we also find that the implicit social discount function is rarely monotone: only 33% of subjects are monotone social discounters or magnifiers, while 27% have a U-shaped and 37% an inverse-U-shaped discount function, placing either the least or the most weight on the intermediate recipient.

**Table G1:** Orderings resulting from $d_{ij}$ estimated.

| Ordering | Bias | Type | Shape | $n$ | % | Cumulative % |
|---|---|---|---|---|---|---|
| $d_{i0} > d_{i3} > d_{i1} > d_{i5}$ | self premium | social discounter | inverse U-shaped | 18 | 19.78 | 19.78 |
| $d_{i0} > d_{i1} > d_{i3} > d_{i5}$ | self premium | social discounter | monotone | 15 | 16.48 | 36.26 |
| $d_{i0} > d_{i3} > d_{i5} > d_{i1}$ | self premium | social magnifier | inverse U-shaped | 13 | 14.29 | 50.55 |
| $d_{i0} > d_{i5} > d_{i3} > d_{i1}$ | self premium | social magnifier | monotone | 12 | 13.19 | 63.74 |
| $d_{i0} > d_{i1} > d_{i5} > d_{i3}$ | self premium | social discounter | U-shaped | 11 | 12.09 | 75.82 |
| $d_{i0} > d_{i5} > d_{i1} > d_{i3}$ | self premium | social magnifier | U-shaped | 10 | 10.99 | 86.81 |
| $d_{i1} > d_{i5} > d_{i0} > d_{i3}$ | other premium | social discounter | U-shaped | 2 | 2.20 | 89.01 |
| $d_{i3} > d_{i0} > d_{i5} > d_{i1}$ | other premium | social magnifier | inverse U-shaped | 2 | 2.20 | 91.21 |
| $d_{i0} > d_{i1} > d_{i3} = d_{i5}$ | self premium | social discounter | | 1 | 1.10 | 92.31 |
| $d_{i0} > d_{i3} > d_{i1} = d_{i5}$ | self premium | | | 1 | 1.10 | 93.41 |
| $d_{i1} > d_{i0} > d_{i3} > d_{i5}$ | other premium | social discounter | monotone | 1 | 1.10 | 94.51 |
| $d_{i1} > d_{i3} > d_{i5} > d_{i0}$ | other premium | social discounter | monotone | 1 | 1.10 | 95.60 |
| $d_{i1} > d_{i5} > d_{i3} > d_{i0}$ | other premium | social discounter | U-shaped | 1 | 1.10 | 96.70 |
| $d_{i3} > d_{i5} > d_{i0} > d_{i1}$ | other premium | social magnifier | inverse U-shaped | 1 | 1.10 | 97.80 |
| $d_{i5} > d_{i0} > d_{i1} > d_{i3}$ | other premium | social magnifier | U-shaped | 1 | 1.10 | 98.90 |
| $d_{i5} > d_{i0} > d_{i3} > d_{i1}$ | other premium | social magnifier | monotone | 1 | 1.10 | 100.00 |
| Total | | | | 91 | 100 | |

*Notes.* $d_{i0}$: weight estimated for the self. $d_{i1}$: for the most similar recipient with known dissimilarity. $d_{i3}$: for the recipient of intermediate known dissimilarity. $d_{i5}$: for the least similar recipient with known dissimilarity. Sample restricted to the 91 subjects with a valid maximum likelihood estimate and complete demographic information for all five recipients.

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